BMJ Open

BMJ Open is committed to open peer review. As part of this commitment we make the peer review history of every article we publish publicly available.

When an article is published we post the peer reviewers' comments and the authors' responses online. We also post the versions of the paper that were used during peer review. These are the versions that the peer review comments apply to.

The versions of the paper that follow are the versions that were submitted during the peer review process. They are not the versions of record or the final published versions. They should not be cited or distributed as the published version of this manuscript.

BMJ Open is an open access journal and the full, final, typeset and author-corrected version of record of the manuscript is available on our site with no access controls, subscription charges or pay-per-view fees (http://bmjopen.bmj.com).

If you have any questions on BMJ Open's open peer review process please email info.bmjopen@bmj.com

BMJ Open

Quality evaluation of clinical practice guidelines for venous thromboembolism prophylaxis in orthopedic trauma based on AGREE II and AGREE-REX:a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-059181
Article Type:	Protocol
Date Submitted by the Author:	10-Nov-2021
Complete List of Authors:	He, Lingxiao; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Chengdu, CN Xie, Jing-ying; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Lv, Juan; Department of Geriatrics and National Clinical Research Center for Geriatrics, West China Hospital, Sichuan University, Chengdu, China Liu, Huan; Department of Critical Care Medicine, West China Hospital, Sichuan University, Chengdu, China Liao, Deng-bin; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Wang, Guang-lin; Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China Ning, Ning; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Zhou, Zong-Ke; Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China
Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Trauma management < ORTHOPAEDIC & TRAUMA SURGERY, VASCULAR MEDICINE

SCHOLARONE™
Manuscripts

Quality evaluation of clinical practice guidelines for venous thromboembolism prophylaxis in orthopedic trauma based on AGREE II and AGREE-REX: a systematic review protocol

Ling-xiao He[1], Jing-ying Xie[1], Juan Lv[2], Huan Liu[3], Deng-bin Liao[1], Guang-lin Wang[4], Ning Ning[1], Zong-ke Zhou[4]

[1] West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China

[2] Department of Geriatrics and National Clinical Research Center for Geriatrics, West China Hospital, Sichuan University, Chengdu, China

[3] Department of Critical Care Medicine, West China Hospital, Sichuan University, Chengdu, China

[4] Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China

* correspondence to: Dr Zong-Ke Zhou and Dr Ning Ning.

ABSTRACT

Introduction Orthopedic trauma patients are at high risk of venous thromboembolism(VTE). As VTE prophylaxis has gradually raised public concern, guidelines relevant to this topic increased over time. However, the recommendations of VTE prophylaxis guidelines in orthopedic trauma patients are inconsistent, and the quality of the guidelines and recommendations for this topic still lacks a comprehensive assessment. This review aims to critically appraise clinical practice guidelines(CPGs) for VTE prophylaxis in orthopedic trauma patients.

Methods and analysis We will conduct a comprehensive literature search up to 31 December 2021 in databases(PubMed, EMBASE, CINAHL, Web of Science, the Cochrane Library, etc.), academic websites, and guidelines repositories. The quality of the guidelines and recommendations will be assessed by five reviewers independently using the Appraisal of Guidelines Research and Evaluation II instrument(AGREE-II) and The Appraisal of Guidelines Research and Evaluation - Recommendation

For peer review only

Excellence(AGREE-REX). We will summarise the characteristics of guidelines and compare the difference between recommendations.

Ethics and dissemination Since this research is based on literature, no ethical consent is required. The results will be summarized as a paper, disseminated through peer-reviewed journals, and help guide further research in the future.

Protocol registration number CRD42021273405.

Keywords orthopedic trauma; venous thromboembolism; clinical practice guideline; systematic review

STRENGTHS AND LIMITATIONS OF THIS STUDY

- We will evaluate the quality of clinical practice guidelines and applicability of the recommendations in guidelines related to VTE prevention in orthopedic trauma patients to guide evidence-based clinical practice.
- We will conduct a comprehensive literature search and use the widely recognized AGREE-II and AGREE-REX tools for rigorous quality evaluation.
- All members participating in the quality evaluation underwent systematic training.
- This study will limit to guidelines written in Chinese and English, which could be a limiting factor.
- We evaluate the methodological quality of CPGs based on the content in the guidelines, which may not be consistent with the actual development process.

INTRODUCTION

Venous Thromboembolism(VTE) is the third most common cause of cardiovascular-related death and results in a mass of disability-adjusted life-years lost in hospitalized patients[1]. Above 500,000 VTE-related deaths happen in the USA[1] and cost $7~10 billion for healthcare annually[2]. As an avoidable in-hospital complication, each case of VTE will take an excess 5.4 hospitalization days and increase the mortality rate by 6.6 times[3]. About 6% of DVT patients and 12% of PE

For peer review only

patients die within one month after diagnosis[4]. A variety of factors can lead to the occurrence of VTE, such as major surgery, trauma, plaster fixation, tumors, etc. [5] Orthopedic trauma patients were at high risk of VTE due to multiple risk factors. The incidence of VTE in orthopedic trauma patients can vary from 4.78% ~ 40.25%[6–9], depending on the location of the injury, VTE monitoring method, ethnicity, etc.

VTE prophylaxis has been proven to be effective in reducing the occurrence of VTE[10]. Both Chemical and mechanical prophylaxis strategies have shown their effects in orthopedic trauma patients. At present, the most widely used VTE prevention clinical practice guideline(CPG) in orthopedic surgery is the 9th edition of the VTE prevention guideline presented by ACCP[11] in 2012. Still, it does not fully include the specific conditions of orthopedic trauma patients. Recently, several updated CPGs for VTE prevention have been published[12–19], but many have been formed based on expert consensus[14,15,18,19]. In addition, there are still many issues to be determined[20], such as the timing, duration, and best practice protocol for VTE prevention of patients with different types of orthopedic trauma. Further constructing a VTE prevention program that fits the clinical situation of orthopedic trauma patients based on guidelines still needs to be explored.

At present, The Appraisal of Guidelines for Research & Evaluation Instrument (AGREE-II) [21] is widely accepted to evaluate the methodological quality of the guideline, but it cannot determine the quality and applicability of the guideline's recommendations. The judgment is not conducive to the choice in clinical practice; The Appraisal of Guidelines Research and Evaluation - Recommendation Excellence （AGREE-REX） tool[22] mainly evaluates the quality and applicability of the recommendations of the guideline, which is a powerful supplement to AGREE-II. We hope to use AGREE-II and AGREE-REX tools to critically appraise the current CPGs for VTE prevention in orthopedic trauma and understand the methodological quality and recommendation quality of the current guidelines to guide selecting reliable recommendations in accordance with the clinical situation.

METHODS AND ANALYSIS

For peer review only

Objectives

We plan to systematically collect VTE prophylaxis CPGs in the field of orthopedic trauma and use AGREE-II and AGREE-REX tools to conduct a rigorous quality appraisal to answer three questions:

1. What is the methodological quality of the VTE prophylaxis CPGs in orthopedic trauma?

2. What similarities and differences are the recommendations provided in the current CPGs for VTE prophylaxis in orthopedic trauma?

3. What is the strength and quality of the evidence provided in the current CPGs for VTE prophylaxis in orthopedic trauma?

Protocol and registration

This report follows the PRISMA-P statement (see appendix 1) and has been registered in the International Prospective Register of Systematic Reviews (PROSPERO) with the registration number CRD42021273405.

Patient and public involvement

Patients and the public did not participate in the design, implementation, or interpretation of the results of this study. However, we will use the results of this study to improve the quality of VTE prevention.

Eligibility criteria

See details in table 1。

Table 1 Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
(1) Guidelines published in English or Chinese; (2) The guideline has clearly stated that it is a "guideline" in the context; (3) The content includes statements and recommendations for the prevention of VTE in orthopedic trauma patients;	(1) Evidence summary, editorials, letters, consensus statements, policy statements or standards, or just a quick reference guide which do not involve the evaluation of evidence and the description of the method of forming recommendations; (2) The previous version of the updated guideline; (3) Repeated publication;

For peer review only

(4) Guidelines published from inception to Dec 2021; (5) The latest complete version of the guideline and any partial revisions or updates published afterward.	(4) Guidelines for a specific institution; (5)Only included one or more of the following: spinal fractures, craniocerebral or maxillofacial fractures, rib fractures, or skin and soft tissue injuries; (6) Unable to get the full text.

SELECTION OF STUDIES

Data sources

We will search the following databases and websites until 31 December, 2021:

(1) Electronic databases: PubMed, EMBASE, CINAHL, Web of Science, the Cochrane Library, SinoMed, CNKI, WanFang.

(2) Academic websites: World Health Organization(WHO, www.who.int), Centers for Disease Control and Prevention(CDC, www.cdc.gov), American Academy of Orthopaedic Surgeons(AAOS, www.aaos.org), American College of Chest Physicians(ACCP, www.chestnet.org), Registered Nurses Association of Ontario(RNAO, rnao.ca), Chinese Medical Association(CMA, www.cma.org.cn)

(3) Guideline repositories: Guidelines International Network (G-I-N), National Institute for Health and Care Excellence(NICE), New Zealand Guidelines Group(NZGG), Scottish Intercollegiate Guidelines Network(SIGN), CPG-Infobase, National Health & Medical Research Council(NHMRC), JBI database, Clinical Key, UpTodate.

(4) others: any additional guidelines identified from references of eligible guidelines.

Search strategies

Search terms related to "orthopedic trauma", "VTE", and "guideline" will be adapted for each database. An example of a database search strategy is provided in Box 1.

Box 1 Sample research strategy for PubMed.

1. "orthopedics"[MeSH Terms] OR "traumatology"[MeSH Terms] OR "wounds and injuries"[MeSH Terms] 2. "fracture*"[Title/Abstract] OR "limb trauma"[Title/Abstract] OR "extremity trauma"[Title/Abstract] OR "musculoskeletal injury"[Title/Abstract] OR "musculoskeletal trauma"[Title/Abstract] OR "limb immobili*"[Title/Abstract] OR ("orthop*"[Title/Abstract] AND "trauma"[Title/Abstract]) OR "skeletal fixation"[Title/Abstract]

For peer review only

3. "venous thromboembolism"[MeSH Terms] OR "venous thrombosis"[MeSH Terms] OR "pulmonary embolism"[MeSH Terms]
4. "venous thromboembolism"[Title/Abstract] OR "venothromboembolism"[Title/Abstract] OR "venous thrombosis"[Title/Abstract] OR "vein thrombosis"[Title/Abstract] OR "DVT"[Title/Abstract] OR "VTE"[Title/Abstract] OR "pulmonary embolism"[Title/Abstract] OR "lung embolism"[Title/Abstract] OR "PE"[Title/Abstract] OR "thromboembolism"[Title/Abstract] OR "embolism"[Title/Abstract] OR "blood clot"[Title/Abstract]
5. "guideline"[Publication Type] OR "consensus development conference"[Publication Type]
6. "guidelines as topic"[MeSH Terms] OR "consensus"[MeSH Terms] OR "consensus development conferences as topic"[MeSH Terms]
7. "guideline"[Title/Abstract] OR "guidance"[Title/Abstract] OR "recommendation"[Title/Abstract] OR "consensus"[Title/Abstract] OR "best practice"[Title/Abstract]
8. 1 OR 2
9. 3 OR 4
10. 5 OR 6 OR 7
11. 8 AND 9 AND 10

Determination of eligibility

Two reviewers(LH, JX) will independently screen and select the literature. First, the reviewer will determine whether the document meets the inclusion criteria based on the title and abstract. After that, the reviewer obtains the full text of the guideline that meets the criteria, reads it, and filters out the eligible documents. Any disagreement between the two reviewers will be resolved through consensus or decided by a third reviewer(ZZ).

DATA EXTRACTION

We will extract the following information of CPGs: (1) title; (2) year of publication; (3) authors; (4) country of development; (5)objective of CPG; (6) method used to collect the evidence; (7) grading methods for the evidence; (8) recommendation formulation method; (9) guidelines validation methods; (10) intended users; (11) recommendations given. In addition, we will record each score of items in AGREE-II and AGREE-REX with their reasons.

For peer review only

QUALITY ASSESSMENT OF CPGS

We will assess the quality of CPGs by AGREE-II[21]. The tool was released in 2009, includes 23 items in 6 domains(*scope and purpose, stakeholder involvement, rigour of development, clarity of presentation, applicability, and editorial independence*), with 2 additional assessments for the guideline's overall quality and whether it is recommended to use.

The quality of recommendations is evaluated using AGREE-REX[22]. The tool was released in 2019 and included 9 items in 3 domains(*clinical applicability, values and preferences, and implementability*). Each item has two assessment questions, one for quality evaluation (required) and another for suitability for use (optional). Overall judgment for all recommendations in CPG have 2 additional assessments: (1) *I would recommend these recommendations for use in the appropriate context*(required); (2) *I would recommend these recommendations for use in my context*(optional).

Both AGREE-II and AGREE-REX quality evaluation items use a 1-7 point Likert scale. The standardized scores for each domain are calculated according to the user manual[23,24], ranging from 0% to 100%. The higher the score obtained, the better the quality of the guideline in this field. Considering that the method for evaluating the overall quality of the guideline remains controversial[25], we do not use a cut-off value to rate the overall quality but directly present the reviewers' subjective judgment results.

Five reviewers conducted quality appraisals after systematically learning how to use AGREE-II and AGREE-REX. Reviewers use the online platform *MY AGREE PLUS* to use AGREE-II independently and blindly (https://www.agreetrust.org/my-agree/), while the AGREE-REX assessment is performed offline.

DESCRIPTION AND COMPARISON OF THE RECOMMENDATIONS

We plan to compare the recommendations included in the CPG, and the recommendations will be extracted independently by two reviewers. The disagreement between the reviewers will be solved through discussion. If the two cannot reach a consensus, another reviewer (ZZ) will draw a decision. It is expected that the recommendations can be classified according to risk assessment, screening strategies,

For peer review only

mechanical prophylaxis, chemical prophylaxis, and others to evaluate the scope and differences of recommendations in CPGs. When recommendations conflict with each other, these recommendations will be highlighted.

DATA MANAGEMENT AND SYNTHESIS

All data are saved using Excel (Microsoft 365, Microsoft Corporation, Redmond, WA, USA) and analyzed using SPSS statistical software (Version 26, IBM Corporation, Armonk, NY, USA). For each of AGREE II and AGREE-REX In the field, we use descriptive statistical methods, and the results are described using Mean (SD) and Median (IQR) (if needed). The intra-class correlation coefficients (ICC) was used to evaluate the degree of agreement between reviewers. Based on Landis and Koch[26], an ICC value of 0.00~0.20 indicates slight agreement, 0.21 ~ 0.40 indicates fair agreement, >0.41 ~ 0.61 indicates moderate agreement, >0.61 ~ 0.80 indicates substantial agreement, 0.81~1.00 indicates almost perfect agreement. Differences in domain scores between CPGs were compared using the Kruskal-Wallis H test. Significant level $\alpha = 0.05$(two-tailed).

DISCUSSION

VTE prophylaxis in orthopedic trauma patients is complicated due to the diversity of risk factors. It is challenging to choose an appropriate prevention method for patients with coagulation abnormalities, lower limb injuries, etc. Research in this field is scattered and controversial, and more research is needed to draw conclusions[27]. There is still no consensus on evaluating evidence and recommendations between different guidelines, which increases the uncertainty of clinical decision-making[28]. Even in the relatively large number of hip fracture patients studied, physicians still believe that there is a lack of adequate clinical guidelines[29]. We hope to evaluate the relevant guidelines for VTE prevention in orthopedic trauma and critically appraise the quality of the guidelines and recommendations. As far as we know, there is currently no systematic review on the quality of English and Chinese orthopedic trauma VTE prevention guidelines.

For peer review only

It's a long process that costs plenty of workforce and resources for planning, formulating, and disseminating a CPG, especially for high-quality CPGs. Rigorous methodology and regular updates play an essential role in maintaining the quality of the guidelines and practical application [30]. Identifying high-quality recommendations will help us choose an appropriate way to intervene target population. It will also help discover knowledge gaps and new points for further research[31].

Our research may still have shortcomings: First, although we have adopted a comprehensive search strategy, we only obtain openly published documents and may miss some grey literature. Second, we include CPGs published in Chinese and English only, which may lead to underestimating the overall number of guidelines. Third, to obtain as comprehensive information as possible, we do not limit the start time for the publication of the guidelines. Those recommendations from earlier CPGs may not be consistent with current evidence. Finally, we are scoring the CPGs based on the content shown in the guideline document, which may not be consistent with the actual development process.

CONTRIBUTORS

Study design: LH, JX, NN and ZZ. Drafting the protocol: LX, JX, JL, HL and DL. Review and editing the manuscript: LX, GW, NN and ZZ. All authors read, discussed and approved the final manuscript. NN and ZZ contributed equally to this paper.

FUNDING

None declared.

COMPETING INTERESTS

None declared.

PATIENT AND PUBLIC INVOLVEMENT

Patients and/or the public were not involved in the design, conduct, reporting, or dissemination plans of this research.

PATIENT CONSENT FOR PUBLICATION

Not required.

ETHICS APPROVAL

For peer review only

Not required.

REFERENCES

1 Raskob GE, Angchaisuksiri P, Blanco AN, *et al.* Thrombosis: a major contributor to global disease burden. *Arterioscler Thromb Vasc Biol* 2014;**34**:2363–71. doi:10.1161/ATVBAHA.114.304488

2 Grosse SD, Nelson RE, Nyarko KA, *et al.* The economic burden of incident venous thromboembolism in the United States: A review of estimated attributable healthcare costs. *Thromb Res* 2016;**137**:3–10. doi:10.1016/j.thromres.2015.11.033

3 Zhan C. Excess Length of Stay, Charges, and Mortality Attributable to Medical Injuries During Hospitalization. *JAMA* 2003;**290**:1868. doi:10.1001/jama.290.14.1868

4 White RH. The epidemiology of venous thromboembolism. *Circulation* 2003;**107**:I4-8. doi:10.1161/01.CIR.0000078468.11849.66

5 Wolberg AS, Rosendaal FR, Weitz JI, *et al.* Venous thrombosis. *Nat Rev Dis Primers* 2015;**1**:15006. doi:10.1038/nrdp.2015.6

6 Wu L, Cheng B. Analysis of perioperative risk factors for deep vein thrombosis in patients with femoral and pelvic fractures. *J Orthop Surg Res* 2020;**15**:597. doi:10.1186/s13018-020-02131-5

7 Fan J, Zhou F, Xu X, *et al.* Clinical predictors for deep vein thrombosis on admission in patients with intertrochanteric fractures: a retrospective study. *BMC Musculoskelet Disord* 2021;**22**:328. doi:10/gkrb59

8 Kanchanabat B, Stapanavatr W, Meknavin S, *et al.* Systematic review and meta-analysis on the rate of postoperative venous thromboembolism in orthopaedic surgery in Asian patients without thromboprophylaxis. *Br J Surg* 2011;**98**:1356–64. doi:10.1002/bjs.7589

9 Yumoto T, Naito H, Yamakawa Y, *et al.* Venous thromboembolism in major trauma patients: a single-center retrospective cohort study of the epidemiology and utility of D-dimer for screening. *Acute Med Surg* 2017;**4**:394–400. doi:10.1002/ams2.290

10 Barrera LM, Perel P, Ker K, *et al.* Thromboprophylaxis for trauma patients. *Cochrane Database Syst Rev* 2013;:CD008303. doi:10.1002/14651858.CD008303.pub2

11 Falck-Ytter Y, Francis CW, Johanson NA, *et al.* Prevention of VTE in Orthopedic Surgery Patients. Antithrombotic Therapy and Prevention of Thrombosis, 9th ed:

For peer review only

American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest* 2012;**141**:e278S-e325S. doi:10.1378/chest.11-2404

12 National Institute for Health and Care Excellence. Venous thromboembolism in over 16s: reducing the risk of hospital-acquired deep vein thrombosis or pulmonary embolism. National Institute for Health and Care Excellence. 2019.https://www.nice.org.uk/guidance/ng89 (accessed 27 Jun 2021).

13 Anderson DR, Morgano GP, Bennett C, *et al.* American Society of Hematology 2019 guidelines for management of venous thromboembolism: prevention of venous thromboembolism in surgical hospitalized patients. *Blood Adv* 2019;**3**:3898–944. doi:10.1182/bloodadvances.2019000975

14 Ngarmukos S, Kim K-I, Wongsak S, *et al.* Asia-Pacific venous thromboembolism consensus in knee and hip arthroplasty and hip fracture surgery: Part 1. Diagnosis and risk factors. *Knee Surg & Relat Res* 2021;**33**:18. doi:10.1186/s43019-021-00099-y

15 Amarase C, Tanavalee A, Larbpaiboonpong V, *et al.* Asia-Pacific venous thromboembolism consensus in knee and hip arthroplasty and hip fracture surgery: Part 2. Mechanical venous thromboembolism prophylaxis. *Knee Surg Relat Res* 2021;**33**:20. doi:10.1186/s43019-021-00101-7

16 Afshari A, Ageno W, Ahmed A, *et al.* European Guidelines on perioperative venous thromboembolism prophylaxis: Executive summary. *Eur J Anaesthesiol* 2018;**35**:77–83. doi:10.1097/EJA.0000000000000729

17 Aggarwal S, Patel S, Vashisht S, *et al.* Guidelines for the prevention of venous thromboembolism in hospitalized patients with pelvi-acetabular trauma. *J Clin Orthop Trauma* 2020;**11**:1002–8. doi:10.1016/j.jcot.2020.09.011

18 Lin Q, Yang M, Hou Z. Guidelines for prevention of perioperative venous thromboembolism in Chinese orthopedic trauma patients(2021). *Chin J Orthop Trauma* 2021;**23**:185–92. doi:10.3760/cma.j.cn115530-20201228-00795.

19 Ley EJ, Brown CVR, Moore EE, *et al.* Updated guidelines to reduce venous thromboembolism in trauma patients: A Western Trauma Association critical decisions algorithm. *J Trauma Acute Care Surg* 2020;**89**:971–81. doi:10.1097/TA.0000000000002830

20 Sagi HC, Ahn J, Ciesla D, *et al.* Venous Thromboembolism Prophylaxis in Orthopaedic Trauma Patients: A Survey of OTA Member Practice Patterns and OTA Expert Panel Recommendations. *J Orthop Trauma* 2015;**29**:8. doi:10.1097/BOT.0000000000000387

21 Brouwers MC, Kho ME, Browman GP, *et al.* AGREE II: advancing guideline development, reporting and evaluation in health care. *CMAJ* 2010;**182**:E839-842.

For peer review only

doi:10.1503/cmaj.090449

22 Brouwers MC, Spithoff K, Kerkvliet K, *et al.* Development and Validation of a Tool to Assess the Quality of Clinical Practice Guideline Recommendations. *JAMA Network Open* 2020;**3**:e205535–e205535. doi:10.1001/jamanetworkopen.2020.5535

23 AGREE Next Steps Consortium. The AGREE II Instrument. 2017.https://www.agreetrust.org/resource-centre/agree-ii/ (accessed 19 Oct 2021).

24 AGREE–REX Research Team. The Appraisal of Guidelines Research & Evaluation—Recommendation EXcellence (AGREE-REX). 2019.https://www.agreetrust.org/resource-centre/agree-rex-recommendation-excellence/ (accessed 19 Oct 2021).

25 Bargeri S, Iannicelli V, Castellini G, *et al.* AGREE II appraisals of clinical practice guidelines in rehabilitation showed poor reporting and moderate variability in quality ratings when users apply different cuff-offs: a methodological study. *J Clin Epidemiol* 2021;**139**:222–31. doi:10.1016/j.jclinepi.2021.08.021

26 Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics* 1977;**33**:159–74.

27 Flevas DA, Megaloikonomos PD, Dimopoulos L, *et al.* Thromboembolism prophylaxis in orthopaedics: an update. *EFORT Open Rev* 2018;**3**:136–48. doi:10.1302/2058-5241.3.170018

28 Gesu E, Bellone P, Bonzi M, *et al.* Management of patients with cervical spine trauma in the emergency department: a systematic critical appraisal of guidelines with a view to developing standardized strategies for clinical practice. *Intern Emerg Med* Published Online First: 5 October 2021. doi:10.1007/s11739-021-02838-1

29 White NJ, Reitzel SL, Doyle-Baker D, *et al.* Management of patients with hip fracture receiving anticoagulation: What are we doing in Canada? *Can J Surg* 2021;**64**:E510–5. doi:10.1503/cjs.018520

30 Cosgrove L, Shaughnessy AF, Shaneyfelt T. When is a guideline not a guideline? The devil is in the details. *BMJ Evid Based Med* 2018;**23**:33–6. doi:10.1136/ebmed-2017-110845

31 Murad MH. Clinical Practice Guidelines: A Primer on Development and Dissemination. *Mayo Clin Proc* 2017;**92**:423–33. doi:10.1016/j.mayocp.2017.01.001

For peer review only

PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1-3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	35
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	online
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	229-232
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	n/a
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	233-234
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	233-234
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	235-236
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	54-87
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	91-99
METHODS					

For peer review only

BioMed Central The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	111-113
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	116-129
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	132-135
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	187-198
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	138-143
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	187-198
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	146-151
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	154-175
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	154-175
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☐	☒	n/a
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☐	☒	n/a
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☒	n/a
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	188-198
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective reporting within studies)	☐	☒	n/a
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☐	☒	n/a

For peer review only

BioMed Central
The Open Access Publisher

BMJ Open

Quality evaluation of clinical practice guidelines for thromboprophylaxis in orthopedic trauma based on AGREE II and AGREE-REX: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-059181.R1
Article Type:	Protocol
Date Submitted by the Author:	13-Jun-2022
Complete List of Authors:	He, Lingxiao; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Chengdu, CN Xie, Jing-ying; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Lv, Juan; Department of Geriatrics and National Clinical Research Center for Geriatrics, West China Hospital, Sichuan University, Chengdu, China Liu, Huan; Department of Critical Care Medicine, West China Hospital, Sichuan University, Chengdu, China Liao, Deng-bin; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Wang, Guang-lin; Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China Ning, Ning; West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China Zhou, Zong-Ke; Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China
Primary Subject Heading:	Evidence based practice
Secondary Subject Heading:	Nursing, Research methods, Surgery
Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Trauma management < ORTHOPAEDIC & TRAUMA SURGERY, ORTHOPAEDIC & TRAUMA SURGERY

SCHOLARONE™
Manuscripts

Quality evaluation of clinical practice guidelines for thromboprophylaxis in orthopedic trauma based on AGREE II and AGREE-REX: a systematic review protocol

Ling-xiao He[1], Jing-ying Xie[1], Juan Lv[2], Huan Liu[3], Deng-bin Liao[1], Guang-lin Wang[4], Ning Ning[1], Zong-ke Zhou[4]

Author affiliations

[1] West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China

[2] Department of Geriatrics and National Clinical Research Center for Geriatrics, West China Hospital, Sichuan University, Chengdu, China

[3] Department of Critical Care Medicine, West China Hospital, Sichuan University, Chengdu, China

[4] Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China

Correspondence to:

Dr Zong-Ke Zhou: zhouzongke@scu.edu.cn and Dr Ning Ning: gkningning@126.com

postal address: Guoxue Road 37#, Chengdu, Sichuan, China 610041

For peer review only

ABSTRACT

Introduction Orthopedic trauma patients are at high risk of venous thromboembolism (VTE). As VTE prophylaxis has gradually raised public concerns, guidelines related to this topic have increased over time. However, the existing recommendations of thromboprophylaxis guidelines in orthopedic trauma patients are still inconsistent, and the quality of the guidelines and recommendations for the topic still lacks comprehensive assessments. This review aims to critically appraise clinical practice guidelines (CPGs) for thromboprophylaxis in orthopedic trauma patients.

Methods and analysis We will conduct a comprehensive literature search up to 31 December 2021 in databases (PubMed, EMBASE, CINAHL, Web of Science, the Cochrane Library, etc.), academic websites, and guideline repositories. The quality of

the guidelines and recommendations will be assessed by five reviewers independently using the Appraisal of Guidelines Research and Evaluation II instrument (AGREE-II) and the Appraisal of Guidelines Research and Evaluation - Recommendation Excellence (AGREE-REX). We will summarize the characteristics of the guidelines and compare the differences between these recommendations.

Ethics and dissemination This study will follow the Declaration of Helsinki and has received approval from the Ethics Committee on Biomedical Research, West China Hospital, Sichuan University (Ethics Approval No. 2021-989). The results will be summarized as a paper, disseminated through peer-reviewed journals, and will help guide further research in the future.

Protocol registration number CRD42021273405.

Keywords orthopedic trauma; venous thromboembolism; thromboprophylaxis; clinical practice guideline; systematic review

STRENGTHS AND LIMITATIONS OF THIS STUDY

- We will evaluate the quality of clinical practice guidelines and the applicability of the recommendations in guidelines related to VTE prevention in orthopedic trauma patients to guide evidence-based clinical practice.
- We will conduct a comprehensive literature search and use the widely recognized AGREE-II and AGREE-REX tools for rigorous quality evaluation.
- All the members will not participate in the quality appraisal until received systematic review training.
- This study will be limited to guidelines written in Chinese and English, which could be a limiting factor.
- This study might be included in guidelines that were developed earlier or not updated in time. Thus, more cautious interpretations of the study results should be needed.

INTRODUCTION

For peer review only

Venous thromboembolism (VTE), including deep venous thrombosis (DVT) and pulmonary embolism(PE), is the third most common cause of cardiovascular-related death and results in a mass of disability-adjusted life-years lost in hospitalized patients [1]. More than 500,000 VTE-related deaths occur in the USA [1] and cost $7~10 billion for healthcare annually [2]. As an avoidable in-hospital complication, each case of VTE will take an excess of 5.4 hospitalization days and increase the mortality rate by 6.6 times [3]. About 6% of patients with DVT and 12% of patients with PE die within one month after diagnosis [4]. Various factors can lead to the occurrence of VTE, such as major surgery, trauma, plaster fixation, tumors, etc. [5]. Orthopedic trauma patients are at high risk of VTE due to multiple risk factors. The incidence of VTE in orthopedic trauma patients can vary from 4.78% to 40.25% [6–9], depending on the location of the injury, VTE monitoring method, ethnicity, etc.

Thromboprophylaxis has been proven able to reduce the occurrence of VTE effectively [10]. Both chemical and mechanical prophylaxis strategies have shown their effects in orthopedic trauma patients. At present, the most widely used VTE prevention clinical practice guideline (CPG) in orthopedic surgery is the 9th edition of the VTE prevention guideline presented by the American College of Chest Physicians (ACCP) in 2012 [11]. However, its recommendations for patients with orthopedic trauma are insufficient, covering only patients with hip fractures, isolated lower leg injuries, and knee arthroscopy. And no specific recommendations for the dose and frequency of chemoprophylaxis are mentioned in the above ACCP guideline. Recently, several updated CPGs for VTE prevention have been published [12–19], but many have been formed based on expert consensus [14,15,18,19]. In addition, there are still many issues to be determined [20], such as the timing, duration, and best practice protocol for VTE prevention in patients with different types of orthopedic trauma. It is still necessary to explore further construction of a VTE prevention program that fits the clinical situation of orthopedic trauma patients based on guidelines.

Currently, the Appraisal of Guidelines for Research & Evaluation II instrument (AGREE-II) [21] is widely accepted to evaluate the methodological quality of the guidelines. However, this tool does not assess the specific recommendations provided

For peer review only

in the guidelines and cannot guarantee their applicability in clinical practice. The Appraisal of Guidelines Research and Evaluation - Recommendation Excellence (AGREE-REX) tool [22] mainly evaluates the quality and applicability of the guideline recommendations, which is a powerful supplement to AGREE-II. We hope to use the AGREE-II and AGREE-REX tools to critically appraise the current CPGs for VTE prevention in orthopedic trauma and understand both the methodological and recommendation quality of the current guidelines, which help guide the selection of reliable recommendations following the clinical situation.

METHODS AND ANALYSIS

Objectives

We plan to systematically collect VTE prophylaxis CPGs in the field of orthopedic trauma and use AGREE-II and AGREE-REX tools to conduct a rigorous quality appraisal to answer the following questions:

1. What is the methodological quality of the VTE prophylaxis CPGs in orthopedic trauma?
2. What similarities and differences are the recommendations provided in the current CPGs for VTE prophylaxis in orthopedic trauma?
3. What is the strength and quality of the evidence provided in the current CPGs for VTE prophylaxis in orthopedic trauma?

Protocol and registration

This report follows the PRISMA-P statement (see appendix 1) and has been registered in the International Prospective Register of Systematic Reviews (PROSPERO) with the registration number CRD42021273405.

Patient and public involvement

Patients and/or the public will not be involved in the design, conduct, reporting, or dissemination plans of this research.

For peer review only

Eligibility criteria

See details in table 1. We will only include openly published literature in the study and gray literature will be excluded.

Table 1 Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
(1) Guidelines published in English or Chinese; (2) The guideline has clearly stated that it is a "guideline" in the context; (3) The content includes statements and recommendations for the prevention of VTE in orthopedic trauma patients; (4) Guidelines published from inception to Oct 2022; (5) The latest complete version of the guideline and any partial revisions or updates published afterward.	(1) Evidence summary, editorials, letters, consensus statements, policy statements or standards, or just a quick reference guide which do not involve the evaluation of evidence and the description of the method of forming recommendations; (2) The previous version of the updated guideline; (3) Repeated publication; (4) Guidelines for a specific institution; (5) Only included one or more of the following: spinal fractures, craniocerebral or maxillofacial fractures, rib fractures, or skin and soft tissue injuries; (6) Unable to get the full text.

SELECTION OF STUDIES

Data sources

The following databases and websites from inception to 31 October 2022 will be searched:

(1) Electronic databases: PubMed, EMBASE, CINAHL, Web of Science, the Cochrane Library, SinoMed, CNKI, WanFang.

(2) Academic websites: World Health Organization (WHO, www.who.int), Centers for Disease Control and Prevention (CDC, www.cdc.gov), American Academy of Orthopedic Surgeons (AAOS, www.aaos.org), American College of Chest Physicians (ACCP, www.chestnet.org), Registered Nurses Association of Ontario (RNAO, rnao.ca), Chinese Medical Association (CMA, www.cma.org.cn)

(3) Guideline repositories: Guidelines International Network (G-I-N), National Institute for Health and Care Excellence (NICE), New Zealand Guidelines Group (NZGG), Scottish Intercollegiate Guidelines Network (SIGN), CPG-Infobase, National Health & Medical Research Council (NHMRC), JBI database, Clinical Key, UpToDate.

(4) others: any additional guidelines identified from references of eligible guidelines.

For peer review only

Search strategies

Terms related to “orthopedic trauma”, “VTE”, “prevention” and “guideline” will be adopted for the search in each database. Detailed search strategies are listed in appendix 2. An example of a database search strategy is shown in Box 1.

Box 1 Sample research strategy for PubMed.

#1 ((orthopedics[MeSH Terms]) OR (traumatology[MeSH Terms])) OR (wounds and injuries[MeSH Terms])
#2 (((((((fracture[Title/Abstract]) OR (limb trauma[Title/Abstract])) OR (extremity trauma[Title/Abstract])) OR (musculoskeletal injur*[Title/Abstract])) OR (musculoskeletal trauma[Title/Abstract])) OR (limb immobili*[Title/Abstract])) OR (skeletal fixation[Title/Abstract])) OR ("orthop*"[Title/Abstract] AND "trauma"[Title/Abstract])
#3 ((venous thromboembolism[MeSH Terms]) OR (venous thrombosis[MeSH Terms])) OR (pulmonary embolism[MeSH Terms])
#4 ((((((((((venous thromboembolism[Title/Abstract]) OR (venothromboembolism[Title/Abstract])) OR (venous thrombosis[Title/Abstract])) OR (vein thrombosis[Title/Abstract])) OR (DVT[Title/Abstract])) OR (VTE[Title/Abstract])) OR (pulmonary embolism[Title/Abstract])) OR (lung embolism[Title/Abstract])) OR (PE[Title/Abstract])) OR (thromboembolism[Title/Abstract])) OR (embolism[Title/Abstract])) OR (blood clot[Title/Abstract])
#5 ((thromboprophylaxis[Title/Abstract]) OR (prophyla*[Title/Abstract])) OR (prevent*[Title/Abstract])
#6 (((((((heparin[MeSH Terms]) OR (heparin, low-molecular-weight[MeSH Terms])) OR (anticoagulants[MeSH Terms])) OR (warfarin[MeSH Terms])) OR (aspirin[MeSH Terms])) OR (vena cava filters[MeSH Terms])) OR (stockings, compression[MeSH Terms])) OR (intermittent pneumatic compression devices[MeSH Terms])
#7 ((pentasaccharide*[Title/Abstract]) OR (fondaparinux[Title/Abstract])) OR (LMWH[Title/Abstract])
#8 (guideline[Publication Type]) OR (consensus development conference[Publication Type])
#9 ((guidelines as topic[MeSH Terms]) OR (consensus[MeSH Terms])) OR (consensus development conferences as topic[MeSH Terms])
#10 ((((guideline[Title/Abstract]) OR (guidance[Title/Abstract])) OR (recommendation[Title/Abstract])) OR (consensus[Title/Abstract])) OR (best practice[Title/Abstract])
#11 #1 OR #2
#12 #3 OR #4
#13 #5 OR #6 OR #7
#14 #8 OR #9 OR #10
#15 #11 AND #12 AND #13 AND #14

For peer review only

Determination of eligibility

Two reviewers (LH and JX) will independently screen and select the literature. First, they will determine whether the documents meets the inclusion criteria based on the titles and abstracts. After that, the reviewers will obtains the full texts of the documents, then read and filter out the eligible guidelines. Any disagreement between the two reviewers will be resolved through consensus or by the third reviewer (ZZ).

DATA EXTRACTION

We will extract the following information of CPGs: (1) title; (2) year of publication; (3) authors; (4) country of development; (5) objective of CPG; (6) target users; (7) method used to collect the evidence; (8) grading methods for the evidence; (9) recommendation formulation method; (10) guideline validation methods; (11) intended users; (12) the specific content of the recommendations given, including the method of thromboprophylaxis, the timing of initiation, dose and duration, and target patients. In addition, we will record each score of items in AGREE-II and AGREE-REX with their reasons.

QUALITY ASSESSMENT OF CPGS

We will assess the quality of CPGs by AGREE-II [21]. The tool was released in 2009 and includes 23 items in six domains (*scope and purpose, stakeholder involvement, rigour of development, clarity of presentation, applicability, and editorial independence*), with two additional assessments for the guideline's overall quality and whether it is recommended to use.

The quality of recommendations is evaluated using AGREE-REX [22]. The tool was released in 2019 and included nine items in three domains (*clinical applicability, values and preferences, and implementability*). Each item has two assessment questions, one for quality evaluation (required) and another for suitability for use (optional). Overall judgment for all recommendations in CPG has two additional assessments: (1) *I would recommend these recommendations for use in the appropriate context* (required); (2) *I would recommend these recommendations for use in my context*

For peer review only

(optional).

Both AGREE-II and AGREE-REX quality evaluation items adopt a Likert scale of one to seven scores. The standardized scores for each domain are calculated according to the user manual [23,24], ranging from 0% to 100%. The higher the score obtained, the better the quality of the guideline in the relevant field. Considering that the method for evaluating the overall quality of the guideline remains controversial [25], we will not use a cut-off value to rate the overall quality but directly present the reviewers' subjective judgment results.

Five reviewers (LH, JX, JL, HL and NN) conducted quality appraisals after systematically learning how to use AGREE-II and AGREE-REX. Reviewers use the online platform *MY AGREE PLUS* to use AGREE-II independently and blindly (https://www.agreetrust.org/my-agree/), while the AGREE-REX assessment is performed offline.

DESCRIPTION AND COMPARISON OF THE RECOMMENDATIONS

We plan to compare the recommendations included in the CPG, which will be extracted independently by two reviewers (DL and GL). The recommendations will be expectantly classified according to risk assessment, screening strategies, mechanical prophylaxis, chemical prophylaxis, and other interventions to evaluate their scopes and differences in different CPGs. Discrepancies will be highlighted when there are conflicts between some recommendations.

DATA ANALYSIS AND PRESENTATION

All the data will be saved with Excel (Microsoft 365, Microsoft Corporation, Redmond, WA, USA) and analyzed with SPSS statistical software (Version 26, IBM Corporation, Armonk, NY, USA). For each domain of AGREE II and AGREE-REX, We will calculate the score by the following formula:

$$\textit{The scaled domain score} = \frac{\text{Obtained score} - \text{Minimum possible score}}{\text{Maximun possible score} - \text{Minimum possible score}}$$

Scores for guidelines and recommendations will be described using the mean (SD)

For peer review only

and median (IQR) (if needed). The intraclass correlation coefficient (ICC) will be used to evaluate the degree of agreement between reviewers. Based on Landis and Koch [26], an ICC value of 0.00 ~ 0.20 indicates slight agreement, 0.21 ~ 0.40 indicates fair agreement, 0.41 ~ 0.61 indicates moderate agreement, 0.61 ~ 0.80 indicates substantial agreement, 0.81 ~ 1.00 indicates almost perfect agreement. Differences in domain scores between CPGs will be compared using the Kruskal-Wallis H test. Significance level $\alpha = 0.05$ (two-tailed).

Our study is expected to begin in Nov 2022 and complete in April 2023. We will explain any methodological changes in the study. The results are intended to be published in a peer-reviewed open journal.

DISCUSSION

Thromboprophylaxis in orthopedic trauma patients is complicated due diverse risk factors. It is challenging to choose appropriate prevention methods for patients with coagulation abnormalities, lower limb injuries, etc. The use of anticoagulants may increase the risk of bleeding and wound complications in trauma patients [27–29], especially in the early postoperative period. Research in this field is scattered and controversial, and more work is needed to draw conclusions [30]. There is still no consensus on evaluating evidence and recommendations between different guidelines, which increases the uncertainty of clinical decision-making [31]. Even in the relatively large number of hip fracture patients studied, physicians still believe that there is a lack of adequate clinical guidelines [32]. We hope to evaluate the relevant guidelines for VTE prevention in orthopedic trauma and critically appraise the quality of the guidelines and recommendations. To the best of our knowledge, there is currently no systematic review on the quality of English and Chinese orthopedic trauma thromboprophylaxis guidelines.

It is a long process that costs plenty of workforce and resources for planning, formulating, and disseminating a CPG, especially for high-quality CPGs. Rigorous methodology and regular updates play essential roles in maintaining the quality of the guidelines and practical application [33]. Identifying high-quality recommendations

For peer review only

will help us to prescribe appropriate thromboprophylaxis in the trauma population. It will also help discover knowledge gaps and new points for further research [34]. We plan to summarize the divergent recommendations in such guidelines, which will be the main focus of future thromboprophylaxis research.

Our research may still have shortcomings. First, although we will adopted a comprehensive search strategy, only openly published documents will be obtained, and some gray literature may miss during the process. Second, we will only include CPGs published in Chinese and English, which may lead to an underestimation of the overall number of guidelines. Third, to obtain as comprehensive information as possible, we will not limit the start time for the publication of the guidelines. Those recommendations from earlier CPGs may not be consistent with current evidence.

CONTRIBUTORS

Study design: LH, JX, NN and ZZ. Protocol drafting: LX, JX, JL, HL and DL. Manuscript reviewing and editing: LX, GW, NN and ZZ. All the authors have read, discussed, and approved the final version of the manuscript. NN and ZZ contributed equally to the manuscript.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the suggestions given by Mr. Lei Luo on the manuscript language editing.

FUNDING

None declared.

COMPETING INTERESTS

None declared.

PATIENT AND PUBLIC INVOLVEMENT

Patients and/or the public will not be involved in the design, conduct, reporting, or dissemination plans of this research.

ETHICS AND DISSEMINATION

This study will follow the Declaration of Helsinki and has received approval from the Ethics Committee on Biomedical Research, West China Hospital, Sichuan University

For peer review only

(Ethics Approval No. 2021-989). The results will be summarized as a paper, disseminated through peer-reviewed journals, and will help guide further research in the future.

PATIENT CONSENT FOR PUBLICATION

Not required.

REFERENCES

1 Raskob GE, Angchaisuksiri P, Blanco AN, *et al.* Thrombosis: a major contributor to global disease burden. *Arterioscler Thromb Vasc Biol* 2014;**34**:2363–71. doi:10.1161/ATVBAHA.114.304488

2 Grosse SD, Nelson RE, Nyarko KA, *et al.* The economic burden of incident venous thromboembolism in the United States: A review of estimated attributable healthcare costs. *Thromb Res* 2016;**137**:3–10. doi:10.1016/j.thromres.2015.11.033

3 Zhan C. Excess Length of Stay, Charges, and Mortality Attributable to Medical Injuries During Hospitalization. *JAMA* 2003;**290**:1868. doi:10.1001/jama.290.14.1868

4 White RH. The epidemiology of venous thromboembolism. *Circulation* 2003;**107**:I4-8. doi:10.1161/01.CIR.0000078468.11849.66

5 Wolberg AS, Rosendaal FR, Weitz JI, *et al.* Venous thrombosis. *Nat Rev Dis Primers* 2015;**1**:15006. doi:10.1038/nrdp.2015.6

6 Wu L, Cheng B. Analysis of perioperative risk factors for deep vein thrombosis in patients with femoral and pelvic fractures. *J Orthop Surg Res* 2020;**15**:597. doi:10.1186/s13018-020-02131-5

7 Fan J, Zhou F, Xu X, *et al.* Clinical predictors for deep vein thrombosis on admission in patients with intertrochanteric fractures: a retrospective study. *BMC Musculoskelet Disord* 2021;**22**:328. doi:10/gkrb59

8 Kanchanabat B, Stapanavatr W, Meknavin S, *et al.* Systematic review and meta-analysis on the rate of postoperative venous thromboembolism in orthopaedic surgery in Asian patients without thromboprophylaxis. *Br J Surg* 2011;**98**:1356–64. doi:10.1002/bjs.7589

9 Yumoto T, Naito H, Yamakawa Y, *et al.* Venous thromboembolism in major trauma patients: a single-center retrospective cohort study of the epidemiology and utility of D-dimer for screening. *Acute Med Surg* 2017;**4**:394–400.

For peer review only

doi:10.1002/ams2.290

10 Barrera LM, Perel P, Ker K, *et al.* Thromboprophylaxis for trauma patients. *Cochrane Database Syst Rev* 2013;:CD008303. doi:10.1002/14651858.CD008303.pub2

11 Falck-Ytter Y, Francis CW, Johanson NA, *et al.* Prevention of VTE in Orthopedic Surgery Patients. Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest* 2012;**141**:e278S-e325S. doi:10.1378/chest.11-2404

12 National Institute for Health and Care Excellence. Venous thromboembolism in over 16s: reducing the risk of hospital-acquired deep vein thrombosis or pulmonary embolism. National Institute for Health and Care Excellence. 2019.https://www.nice.org.uk/guidance/ng89 (accessed 27 Jun 2021).

13 Anderson DR, Morgano GP, Bennett C, *et al.* American Society of Hematology 2019 guidelines for management of venous thromboembolism: prevention of venous thromboembolism in surgical hospitalized patients. *Blood Adv* 2019;**3**:3898–944. doi:10.1182/bloodadvances.2019000975

14 Ngarmukos S, Kim K-I, Wongsak S, *et al.* Asia-Pacific venous thromboembolism consensus in knee and hip arthroplasty and hip fracture surgery: Part 1. Diagnosis and risk factors. *Knee Surg & Relat Res* 2021;**33**:18. doi:10.1186/s43019-021-00099-y

15 Amarase C, Tanavalee A, Larbpaiboonpong V, *et al.* Asia-Pacific venous thromboembolism consensus in knee and hip arthroplasty and hip fracture surgery: Part 2. Mechanical venous thromboembolism prophylaxis. *Knee Surg Relat Res* 2021;**33**:20. doi:10.1186/s43019-021-00101-7

16 Afshari A, Ageno W, Ahmed A, *et al.* European Guidelines on perioperative venous thromboembolism prophylaxis: Executive summary. *Eur J Anaesthesiol* 2018;**35**:77–83. doi:10.1097/EJA.0000000000000729

17 Aggarwal S, Patel S, Vashisht S, *et al.* Guidelines for the prevention of venous thromboembolism in hospitalized patients with pelvi-acetabular trauma. *Journal of Clinical Orthopaedics and Trauma* 2020;**11**:1002–8. doi:10.1016/j.jcot.2020.09.011

18 Lin Q, Yang M, Hou Z. Guidelines for prevention of perioperative venous thromboembolism in Chinese orthopedic trauma patients(2021). Chin J Orthop Trauma 2021;23:185–92. doi:10.3760/cma.j.cn115530-20201228-00795.

19 Ley EJ, Brown CVR, Moore EE, *et al.* Updated guidelines to reduce venous thromboembolism in trauma patients: A Western Trauma Association critical decisions algorithm. *J Trauma Acute Care Surg* 2020;**89**:971–81.

For peer review only

doi:10.1097/TA.0000000000002830

20 Sagi HC, Ahn J, Ciesla D, *et al.* Venous Thromboembolism Prophylaxis in Orthopaedic Trauma Patients: A Survey of OTA Member Practice Patterns and OTA Expert Panel Recommendations. *J Orthop Trauma* 2015;**29**:8. doi:10.1097/BOT.0000000000000387

21 Brouwers MC, Kho ME, Browman GP, *et al.* AGREE II: advancing guideline development, reporting and evaluation in health care. *CMAJ* 2010;**182**:E839-842. doi:10.1503/cmaj.090449

22 Brouwers MC, Spithoff K, Kerkvliet K, *et al.* Development and Validation of a Tool to Assess the Quality of Clinical Practice Guideline Recommendations. *JAMA Network Open* 2020;**3**:e205535–e205535. doi:10.1001/jamanetworkopen.2020.5535

23 AGREE Next Steps Consortium. The AGREE II Instrument. 2017.https://www.agreetrust.org/resource-centre/agree-ii/ (accessed 19 Oct 2021).

24 AGREE–REX Research Team. The Appraisal of Guidelines Research & Evaluation—Recommendation EXcellence (AGREE-REX). 2019.https://www.agreetrust.org/resource-centre/agree-rex-recommendation-excellence/ (accessed 19 Oct 2021).

25 Bargeri S, Iannicelli V, Castellini G, *et al.* AGREE II appraisals of clinical practice guidelines in rehabilitation showed poor reporting and moderate variability in quality ratings when users apply different cuff-offs: a methodological study. *Journal of Clinical Epidemiology* 2021;**139**:222–31. doi:10.1016/j.jclinepi.2021.08.021

26 Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics* 1977;**33**:159–74.

27 Villarreal JV, Shibuya N, Jupiter DC. Thromboprophylaxis and Bleeding Complications in Orthopedic and Trauma Patients: A Systematic Review. *The Journal of Foot and Ankle Surgery* 2021;**60**:1014–22. doi:10.1053/j.jfas.2021.03.010

28 Heijboer RRO, Lubberts B, Guss D, *et al.* Venous Thromboembolism and Bleeding Adverse Events in Lower Leg, Ankle, and Foot Orthopaedic Surgery with and without Anticoagulants. *J Bone Joint Surg Am* 2019;**101**:539–46. doi:10.2106/JBJS.18.00346

29 Jones CW, Spasojevic S, Goh G, *et al.* Wound Discharge After Pharmacological Thromboprophylaxis in Lower Limb Arthroplasty. *J Arthroplasty* 2018;**33**:224–9. doi:10.1016/j.arth.2017.07.046

For peer review only

30 Flevas DA, Megaloikonomos PD, Dimopoulos L, *et al.* Thromboembolism prophylaxis in orthopaedics: an update. *EFORT Open Rev* 2018;**3**:136–48. doi:10.1302/2058-5241.3.170018

31 Gesu E, Bellone P, Bonzi M, *et al.* Management of patients with cervical spine trauma in the emergency department: a systematic critical appraisal of guidelines with a view to developing standardized strategies for clinical practice. *Intern Emerg Med* Published Online First: 5 October 2021. doi:10.1007/s11739-021-02838-1

32 White NJ, Reitzel SL, Doyle-Baker D, *et al.* Management of patients with hip fracture receiving anticoagulation: What are we doing in Canada? *Can J Surg* 2021;**64**:E510–5. doi:10.1503/cjs.018520

33 Cosgrove L, Shaughnessy AF, Shaneyfelt T. When is a guideline not a guideline? The devil is in the details. *BMJ Evidence-Based Medicine* 2018;**23**:33–6. doi:10.1136/ebmed-2017-110845

34 Murad MH. Clinical Practice Guidelines: A Primer on Development and Dissemination. *Mayo Clinic Proceedings* 2017;**92**:423–33. doi:10.1016/j.mayocp.2017.01.001

For peer review only

PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1-3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	41
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	online
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	255-258
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	n/a
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	262-263
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	262-263
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	262-263
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	60-98
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	101-110

For peer review only

BioMed Central The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	121-124
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	127-141
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	143-147
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	203-219
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	149-154
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	156-164
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	156-164
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	166-171
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	172–186
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☐	☒	n/a
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☐	☒	n/a
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☒	n/a
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	193-201
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective	☐	☒	n/a

For peer review only

BioMed Central
The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
		reporting within studies)			
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☐	☒	n/a

For peer review only

BioMed Central The Open Access Publisher

Table S1. Search strategy for databases, registries and websites

Database	Search terms
PubMed	#1 ((orthopedics[MeSH Terms]) OR (traumatology[MeSH Terms])) OR (wounds and injuries[MeSH Terms]) #2 (((((((fracture[Title/Abstract]) OR (limb trauma[Title/Abstract])) OR (extremity trauma[Title/Abstract])) OR (musculoskeletal injur*[Title/Abstract])) OR (musculoskeletal trauma[Title/Abstract])) OR (limb immobili*[Title/Abstract])) OR (skeletal fixation[Title/Abstract])) OR ("orthop*"[Title/Abstract] AND "trauma"[Title/Abstract]) #3 ((venous thromboembolism[MeSH Terms]) OR (venous thrombosis[MeSH Terms])) OR (pulmonary embolism[MeSH Terms]) #4 ((((((((((venous thromboembolism[Title/Abstract]) OR (venothromboembolism[Title/Abstract])) OR (venous thrombosis[Title/Abstract])) OR (vein thrombosis[Title/Abstract])) OR (DVT[Title/Abstract])) OR (VTE[Title/Abstract])) OR (pulmonary embolism[Title/Abstract])) OR (lung embolism[Title/Abstract])) OR (PE[Title/Abstract])) OR (thromboembolism[Title/Abstract])) OR (embolism[Title/Abstract])) OR (blood clot[Title/Abstract]) #5 ((thromboprophylaxis[Title/Abstract]) OR (prophyla*[Title/Abstract])) OR (prevent*[Title/Abstract]) #6 (((((((heparin[MeSH Terms]) OR (heparin, low-molecular-weight[MeSH Terms])) OR (anticoagulants[MeSH Terms])) OR (warfarin[MeSH Terms])) OR (aspirin[MeSH Terms])) OR (vena cava filters[MeSH Terms])) OR (stockings, compression[MeSH Terms])) OR (intermittent pneumatic compression devices[MeSH Terms]) #7 ((pentasaccharide*[Title/Abstract]) OR (fondaparinux[Title/Abstract])) OR (LMWH[Title/Abstract]) #8 (guideline[Publication Type]) OR (consensus development conference[Publication Type]) #9 ((guidelines as topic[MeSH Terms]) OR (consensus[MeSH Terms])) OR (consensus development conferences as topic[MeSH Terms]) #10 ((((guideline[Title/Abstract]) OR (guidance[Title/Abstract])) OR (recommendation[Title/Abstract])) OR (consensus[Title/Abstract])) OR (best practice[Title/Abstract]) #11 #1 OR #2 #12 #3 OR #4 #13 #5 OR #6 OR #7 #14 #8 OR #9 OR #10 #15 #11 AND #12 AND #13 AND #14
	1 exp orthopedics/ 2 exp traumatology/ 3 exp musculoskeletal injury/ 4 fracture.ab,ti. 5 limb trauma.ab,ti. 6 extremity trauma.ab,ti. 7 musculoskeletal injur$.ab,ti. 8 musculoskeletal trauma.ab,ti. 9 limb immobili$.ab,ti. 10 skeletal fixation.ab,ti. 11 (orthop$ adj3 trauma).ab,ti. 12 exp venous thromboembolism/ 13 exp lung embolism/ 14 venous thromboembolism.ab,ti. 15 venothromboembolism.ab,ti.

For peer review only

	16 venous thrombosis.ab,ti. 17 vein thrombosis.ab,ti. 18 DVT.ab,ti. 19 VTE.ab,ti. 20 pulmonary embolism.ab,ti. 21 lung embolism.ab,ti. 22 PE.ab,ti. 23 thromboembolism.ab,ti. 24 embolism.ab,ti. 25 blood clot.ab,ti. 26 exp heparin/ or exp low molecular weight heparin/ 27 exp anticoagulant agent/ 28 exp warfarin/ 29 exp acetylsalicylic acid/ 30 exp vena cava filter/ 31 exp compression stocking/ 32 exp intermittent pneumatic compression device/ 33 thromboprophylaxis.ab,ti. 34 prophyla$.ab,ti. 35 prevent$.ab,ti. 36 pentasaccharide$.ab,ti. 37 fondaparinux.ab,ti. 38 LMWH.ab,ti. 39 exp practice guideline/ 40 exp consensus/ or exp consensus development/ 41 guideline.ab,ti. 42 guidance.ab,ti. 43 recommendation.ab,ti. 44 consensus.ab,ti. 45 best practice.ab,ti. 46 1 or 2 or 3 or 4 or 5 or 7 or 8 or 9 or 10 or 11 47 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 48 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 49 39 or 40 or 41 or 42 or 43 or 44 or 45 50 46 and 47 and 48 and 49
EBSCO-CINAHL	S1 MH Orthopedic Surgery OR MH Trauma OR MH (Wounds and Injuries) S2 AB fracture OR AB limb trauma OR AB extremity trauma OR AB musculoskeletal injur* OR AB musculoskeletal trauma OR AB limb immobili* OR AB skeletal fixation OR AB orthop#dic trauma S3 MH Venous Thromboembolism OR MH Pulmonary Embolism S4 AB venous thromboembolism OR AB venothromboembolism OR AB venous thrombosis OR AB vein thrombosis OR AB (DVT OR VTE OR PE) OR AB pulmonary embolism OR AB lung embolism OR AB thromboembolism OR AB embolism OR AB blood clot S5 MH Heparin, Low-Molecular-Weight OR MH Heparin OR MH Anticoagulants OR MH Warfarin OR MH Aspirin OR MH Vena Cava Filters OR MH Compression Garments S6 AB thromboprophylaxis OR AB prophyla* OR AB prevent* OR AB pentassacharide* OR AB fondaparinux OR AB compression stocking* OR AB intermittent pneumatic compression S7 MH Practice Guidelines OR MH Consensus S8 AB guideline OR AB guidance OR AB consensus OR AB recommendation OR AB best practice

For peer review only

	S9 S1 OR S2 S10 S3 OR S4 S11 S5 OR S6 S12 S7 OR S8 S13 S9 AND S10 AND S11 AND S12
Web of Science	1 TS=(fracture OR "limb trauma" OR "extremity trauma" OR "musculoskeletal injur*" OR "musculoskeletal trauma" OR "limb immobili*" OR "skeletal fixation" OR "orthop* NEAR/3 trauma") 2 TS=("venous thromboembolism" OR venothromboembolism OR "venous thrombosis" OR "vein thrombosis" OR DVT OR VTE OR "pulmonary embolism" OR "lung embolism" OR PE OR thromboembolism OR embolism OR "blood clot") 3 TS=(thromboprophylaxis OR prophyla* OR prevent* OR heparin OR LMWH OR anticoagulants OR warfarin OR aspirin OR "vena cava filters" OR "compression stocking" OR "intermittent pneumatic compression devices" OR IPC OR pentassacharide* OR fondaparinux) 4 TS=(guideline OR guidance OR consensus OR recommendation OR "best practice") 5 #4 AND #3 AND #2 AND #1
Cochrane Library(via Ovid)	1 exp orthopedics/ 2 exp traumatology/ 3 exp "wounds and injuries"/ 4 fracture.ab,ti. 5 limb trauma.ab,ti. 6 extremity trauma.ab,ti. 7 musculoskeletal injur$.ab,ti. 8 musculoskeletal trauma.ab,ti. 9 limb immobili$.ab,ti. 10 skeletal fixation.ab,ti. 11 (orthop$ adj3 trauma).ab,ti. 12 exp venous thromboembolism/ 13 exp venous thrombosis/ 14 exp pulmonary embolism/ 15 venous thromboembolism.ab,ti. 16 venothromboembolism.ab,ti. 17 venous thrombosis.ab,ti. 18 vein thrombosis.ab,ti. 19 DVT.ab,ti. 20 VTE.ab,ti. 21 pulmonary embolism.ab,ti. 22 lung embolism.ab,ti. 23 PE.ab,ti. 24 thromboembolism.ab,ti. 25 embolism.ab,ti. 26 blood clot.ab,ti. 27 exp heparin/ 28 exp heparin, low-molecular-weight/ 29 exp anticoagulants/ 30 exp warfarin/ 31 exp aspirin/ 32 exp vena cava filters/ 33 exp stockings, compression/ 34 exp intermittent pneumatic compression devices/ 35 thromboprophylaxis.ab,ti. 36 prophyla$.ab,ti.

For peer review only

	37 prevent$.ab,ti. 38 pentassacharide$.ab,ti. 39 fondaparinux.ab,ti. 40 LMWH.ab,ti. 41 exp guidelines as topic/ 42 exp consensus development conferences as topic/ 43 exp consensus/ 44 guideline.ab,ti. 45 guidance.ab,ti. 46 recommendation.ab,ti. 47 consensus.ab,ti. 48 best practice.ab,ti. 49 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 50 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 51 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 52 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 53 49 and 50 and 51 and 52
SinoMed	1 (("矫形外科学"[加权:扩展] OR "创伤和损伤"[加权:扩展] OR "创伤学"[加权:扩展]) OR("骨折"[常用字段] OR "创伤"[常用字段] OR "肌肉骨骼损伤"[常用字段] OR "肢体制动"[常用字段] OR "内固定"[常用字段] OR "外固定"[常用字段])) AND (人类[特征词]) 2 (("静脉血栓形成"[加权:扩展] OR "静脉血栓栓塞"[加权:扩展] OR "肺栓塞"[加权:扩展]) OR("静脉血栓"[常用字段] OR "肺栓塞"[常用字段] OR "肺梗死"[常用字段] OR "DVT"[常用字段] OR "VTE"[常用字段] OR "PE"[常用字段])) AND (人类[特征词]) 3 (("肝素,低分子量"[加权:扩展] OR "阿司匹林"[加权:扩展] OR "华法林"[加权:扩展] OR "抗凝药"[加权:扩展] OR "化学预防"[加权:扩展] OR "腔静脉滤器"[加权:扩展] OR "间歇性气体压缩装置"[加权:扩展] OR "长袜,压力"[加权:扩展]) OR("化学预防"[常用字段] OR "血栓预防"[常用字段] OR "机械预防"[常用字段] OR "弹力袜"[常用字段] OR "间歇加压装置"[常用字段] OR "LMWH"[常用字段] OR "戊糖"[常用字段] OR "磺达肝癸"[常用字段])) AND (人类[特征词]) 4 (("准则"[加权:扩展] OR "多数赞同"[加权:扩展] OR "总结性报告"[加权:扩展]) OR("指南"[常用字段] OR "共识"[常用字段] OR "规范"[常用字段] OR "指引"[常用字段] OR "专家意见"[常用字段])) AND (人类[特征词]) 5 (#1) AND (#2) AND (#3) AND (#4)
CNKI	SU=‘骨折 + 骨科 + 创伤 + 肌肉骨骼损伤 + 内固定 + 外固定’ AND SU=‘静脉血栓 + 肺栓塞 + 肺梗死 + DVT + VTE + PE’ AND SU=‘肝素 + 低分子肝素 + 阿司匹林 + 华法林 + 抗凝药 + 血栓预防 + 化学预防 + 机械预防 + 腔静脉滤器 + 间歇充气加压 + 弹力袜 + 间歇加压装置 + LMWH + 戊糖 + 磺达肝癸’ AND SU=‘指南 + 共识 + 指引 + 专家意见 + 规范’
WanFang	主题:(骨折 OR 骨科 OR 创伤 OR 肌肉骨骼损伤 OR 内固定 OR 外固定) and 主题:(静脉血栓 OR 肺栓塞 OR 肺梗死 OR DVT OR VTE OR PE) and 主题:(肝素 OR 低分子肝素 OR 阿司匹林 OR 华法林 OR 抗凝药 OR 血栓预防 OR 化学预防 OR 机械预防 OR 腔静脉滤器 OR 间歇充气加压 OR 弹力袜 OR LMWH OR 戊糖 OR 磺达肝癸) and 主题:(指南 OR 指引 OR 共识 OR 专家意见 OR 规范)
Academic Websites	thromboembolism OR deep vein thrombosis OR pulmonary embolism OR VTE OR 静脉血栓 OR 肺栓塞 OR 肺梗死

For peer review only

Guideline repositories	thromboembolism OR deep vein thrombosis OR pulmonary embolism OR VTE OR 静脉血栓 OR 肺栓塞 OR 肺梗死

For peer review only

BMJ Open

Quality evaluation of clinical practice guidelines for thromboprophylaxis in orthopedic trauma based on AGREE II and AGREE-REX: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2021-059181.R2
Article Type:	Protocol
Date Submitted by the Author:	19-Oct-2022
Complete List of Authors:	He, Lingxiao; Sichuan University West China Hospital School of Nursing; Sichuan University West China Hospital Department of Orthopedic Surgery Xie, Jing-ying; Sichuan University West China Hospital School of Nursing; Sichuan University West China Hospital Department of Orthopedic Surgery Lv, Juan; Sichuan University West China Hospital Center of Gerontology and Geriatrics Liu, Huan; Sichuan University West China Hospital Department of Critical Care Medicine Liao, Deng-bin; Sichuan University West China Hospital School of Nursing; Sichuan University West China Hospital Department of Orthopedic Surgery Wang, Guang-lin; Sichuan University West China Hospital Department of Orthopedic Surgery Ning, Ning; Sichuan University West China Hospital School of Nursing; Sichuan University West China Hospital Department of Orthopedic Surgery Zhou, Zong-Ke; Sichuan University West China School of Medicine; Sichuan University West China Hospital Department of Orthopedic Surgery
Primary Subject Heading:	Evidence based practice
Secondary Subject Heading:	Nursing, Research methods, Surgery
Keywords:	Protocols & guidelines < HEALTH SERVICES ADMINISTRATION & MANAGEMENT, Trauma management < ORTHOPAEDIC & TRAUMA SURGERY, ORTHOPAEDIC & TRAUMA SURGERY

SCHOLARONE™
Manuscripts

Quality evaluation of clinical practice guidelines for thromboprophylaxis in orthopedic trauma based on AGREE II and AGREE-REX: a systematic review protocol

Ling-xiao He[1], Jing-ying Xie[1], Juan Lv[2], Huan Liu[3], Deng-bin Liao[1], Guang-lin Wang[4], Ning Ning[1], Zong-ke Zhou[4]

Author affiliations

[1] West China School of Nursing, Sichuan University/Department of Orthopedics, West China Hospital, Sichuan University, Chengdu, China

[2] Department of Geriatrics and National Clinical Research Center for Geriatrics, West China Hospital, Sichuan University, Chengdu, China

[3] Department of Critical Care Medicine, West China Hospital, Sichuan University, Chengdu, China

[4] Department of Orthopedics, West China Hospital/West China School of Medicine, Sichuan University, Chengdu, China

Correspondence to:

Dr Zong-Ke Zhou: zhouzongke@scu.edu.cn and Dr Ning Ning: gkningning@126.com

postal address: Guoxue Road 37#, Chengdu, Sichuan, China 610041

For peer review only

ABSTRACT

Introduction Orthopedic trauma patients are at high risk of venous thromboembolism (VTE). As VTE prophylaxis has gradually raised public concerns, guidelines related to this topic have increased over time. However, the existing recommendations of thromboprophylaxis guidelines in orthopedic trauma patients are still inconsistent, and the quality of the guidelines and recommendations for the topic still lacks comprehensive assessments. This review aims to critically appraise clinical practice guidelines (CPGs) for thromboprophylaxis in orthopedic trauma patients.

Methods and analysis We will conduct a comprehensive literature search up to 31 October 2022 in databases (PubMed, EMBASE, CINAHL, Web of Science, the Cochrane Library, etc.), academic websites, and guideline repositories. The quality of

the guidelines and recommendations will be assessed by five reviewers independently using the Appraisal of Guidelines Research and Evaluation II instrument (AGREE-II) and the Appraisal of Guidelines Research and Evaluation - Recommendation Excellence (AGREE-REX). We will summarize the characteristics of the guidelines and compare the differences between these recommendations.

Ethics and dissemination This study will follow the Declaration of Helsinki and has received approval from the Ethics Committee on Biomedical Research, West China Hospital, Sichuan University (Ethics Approval No. 2021-989). The results will be summarized as a paper, disseminated through peer-reviewed journals, and will help guide further research in the future.

Protocol registration number CRD42021273405.

Keywords orthopedic trauma; venous thromboembolism; thromboprophylaxis; clinical practice guideline; systematic review

STRENGTHS AND LIMITATIONS OF THIS STUDY

- We will evaluate the quality of clinical practice guidelines and the applicability of the recommendations in guidelines related to VTE prevention in orthopedic trauma patients to guide evidence-based clinical practice.
- We will conduct a comprehensive literature search and use the widely recognized AGREE-II and AGREE-REX tools for rigorous quality evaluation.
- All the members will not participate in the quality appraisal until received systematic review training.
- This study will be limited to guidelines written in Chinese and English, which could be a limiting factor.
- This study might be included in guidelines that were developed earlier or not updated in time. Thus, more cautious interpretations of the study results should be needed.

INTRODUCTION

For peer review only

Venous thromboembolism (VTE), including deep venous thrombosis (DVT) and pulmonary embolism(PE), is the third most common cause of cardiovascular-related death and results in a mass of disability-adjusted life-years lost in hospitalized patients [1]. More than 500,000 VTE-related deaths occur in the USA [1] and cost $7~10 billion for healthcare annually [2]. As an avoidable in-hospital complication, each case of VTE will take an excess of 5.4 hospitalization days and increase the mortality rate by 6.6 times [3]. About 6% of patients with DVT and 12% of patients with PE die within one month after diagnosis [4]. Various factors can lead to the occurrence of VTE, such as major surgery, trauma, plaster fixation, tumors, etc. [5]. Orthopedic trauma patients are at high risk of VTE due to multiple risk factors. The incidence of VTE in orthopedic trauma patients can vary from 4.78% to 40.25% [6–9], depending on the location of the injury, VTE monitoring method, ethnicity, etc.

Thromboprophylaxis has been proven able to reduce the occurrence of VTE effectively [10]. Both chemical and mechanical prophylaxis strategies have shown their effects in orthopedic trauma patients. At present, the most widely used VTE prevention clinical practice guideline (CPG) in orthopedic surgery is the 9th edition of the VTE prevention guideline presented by the American College of Chest Physicians (ACCP) in 2012 [11]. However, its recommendations for patients with orthopedic trauma are insufficient, covering only patients with hip fractures, isolated lower leg injuries, and knee arthroscopy. And no specific recommendations for the dose and frequency of chemoprophylaxis are mentioned in the above ACCP guideline. Recently, several updated CPGs for VTE prevention have been published [12–19], but many have been formed based on expert consensus [14,15,18,19]. In addition, there are still many issues to be determined [20], such as the timing, duration, and best practice protocol for VTE prevention in patients with different types of orthopedic trauma. It is still necessary to explore further construction of a VTE prevention program that fits the clinical situation of orthopedic trauma patients based on guidelines.

Currently, the Appraisal of Guidelines for Research & Evaluation II instrument (AGREE-II) [21] is widely accepted to evaluate the methodological quality of the guidelines. However, this tool does not assess the specific recommendations provided

For peer review only

in the guidelines and cannot guarantee their applicability in clinical practice. The Appraisal of Guidelines Research and Evaluation - Recommendation Excellence (AGREE-REX) tool [22] mainly evaluates the quality and applicability of the guideline recommendations, which is a powerful supplement to AGREE-II. We hope to use the AGREE-II and AGREE-REX tools to critically appraise the current CPGs for VTE prevention in orthopedic trauma and understand both the methodological and recommendation quality of the current guidelines, which help guide the selection of reliable recommendations following the clinical situation.

METHODS AND ANALYSIS

Objectives

We plan to systematically collect VTE prophylaxis CPGs in the field of orthopedic trauma and use AGREE-II and AGREE-REX tools to conduct a rigorous quality appraisal to answer the following questions:

1. What is the methodological quality of the VTE prophylaxis CPGs in orthopedic trauma?
2. What similarities and differences are the recommendations provided in the current CPGs for VTE prophylaxis in orthopedic trauma?
3. What is the strength and quality of the evidence provided in the current CPGs for VTE prophylaxis in orthopedic trauma?

Protocol and registration

This report follows the PRISMA-P statement (see appendix 1) and has been registered in the International Prospective Register of Systematic Reviews (PROSPERO) with the registration number CRD42021273405.

Patient and public involvement

Patients and/or the public will not be involved in the design, conduct, reporting, or dissemination plans of this research.

For peer review only

Eligibility criteria

See details in table 1. We will only include openly published literature in the study and gray literature will be excluded.

Table 1 Inclusion and exclusion criteria.

Inclusion criteria	Exclusion criteria
(1) Guidelines published in English or Chinese; (2) The guideline has clearly stated that it is a "guideline" in the context; (3) The content includes statements and recommendations for the prevention of VTE in orthopedic trauma patients; (4) Guidelines published from inception to Oct 2022; (5) The latest complete version of the guideline and any partial revisions or updates published afterward.	(1) Evidence summary, editorials, letters, consensus statements, policy statements or standards, or just a quick reference guide which do not involve the evaluation of evidence and the description of the method of forming recommendations; (2) The previous version of the updated guideline; (3) Repeated publication; (4) Guidelines for a specific institution; (5) Only included one or more of the following: spinal fractures, craniocerebral or maxillofacial fractures, rib fractures, or skin and soft tissue injuries; (6) Unable to get the full text.

SELECTION OF STUDIES

Data sources

The following databases and websites from inception to 31 October 2022 will be searched:

(1) Electronic databases: PubMed, EMBASE, CINAHL, Web of Science, the Cochrane Library, SinoMed, CNKI, WanFang.

(2) Academic websites: World Health Organization (WHO, www.who.int), Centers for Disease Control and Prevention (CDC, www.cdc.gov), American Academy of Orthopedic Surgeons (AAOS, www.aaos.org), American College of Chest Physicians (ACCP, www.chestnet.org), Registered Nurses Association of Ontario (RNAO, rnao.ca), Chinese Medical Association (CMA, www.cma.org.cn)

(3) Guideline repositories: Guidelines International Network (G-I-N), National Institute for Health and Care Excellence (NICE), New Zealand Guidelines Group (NZGG), Scottish Intercollegiate Guidelines Network (SIGN), CPG-Infobase, National Health & Medical Research Council (NHMRC), JBI database, Clinical Key, UpToDate.

(4) others: any additional guidelines identified from references of eligible guidelines.

For peer review only

Search strategies

Terms related to “orthopedic trauma”, “VTE”, “prevention” and “guideline” will be adopted for the search in each database. Detailed search strategies are listed in appendix 2. An example of a database search strategy is shown in Box 1.

Box 1 Sample research strategy for PubMed.

#1 ((orthopedics[MeSH Terms]) OR (traumatology[MeSH Terms])) OR (wounds and injuries[MeSH Terms])
#2 (((((((fracture[Title/Abstract]) OR (limb trauma[Title/Abstract])) OR (extremity trauma[Title/Abstract])) OR (musculoskeletal injur*[Title/Abstract])) OR (musculoskeletal trauma[Title/Abstract])) OR (limb immobili*[Title/Abstract])) OR (skeletal fixation[Title/Abstract])) OR ("orthop*"[Title/Abstract] AND "trauma"[Title/Abstract])
#3 ((venous thromboembolism[MeSH Terms]) OR (venous thrombosis[MeSH Terms])) OR (pulmonary embolism[MeSH Terms])
#4 ((((((((((venous thromboembolism[Title/Abstract]) OR (venothromboembolism[Title/Abstract])) OR (venous thrombosis[Title/Abstract])) OR (vein thrombosis[Title/Abstract])) OR (DVT[Title/Abstract])) OR (VTE[Title/Abstract])) OR (pulmonary embolism[Title/Abstract])) OR (lung embolism[Title/Abstract])) OR (PE[Title/Abstract])) OR (thromboembolism[Title/Abstract])) OR (embolism[Title/Abstract])) OR (blood clot[Title/Abstract])
#5 ((thromboprophylaxis[Title/Abstract]) OR (prophyla*[Title/Abstract])) OR (prevent*[Title/Abstract])
#6 (((((((heparin[MeSH Terms]) OR (heparin, low-molecular-weight[MeSH Terms])) OR (anticoagulants[MeSH Terms])) OR (warfarin[MeSH Terms])) OR (aspirin[MeSH Terms])) OR (vena cava filters[MeSH Terms])) OR (stockings, compression[MeSH Terms])) OR (intermittent pneumatic compression devices[MeSH Terms])
#7 ((pentasaccharide*[Title/Abstract]) OR (fondaparinux[Title/Abstract])) OR (LMWH[Title/Abstract])
#8 (guideline[Publication Type]) OR (consensus development conference[Publication Type])
#9 ((guidelines as topic[MeSH Terms]) OR (consensus[MeSH Terms])) OR (consensus development conferences as topic[MeSH Terms])
#10 ((((guideline[Title/Abstract]) OR (guidance[Title/Abstract])) OR (recommendation[Title/Abstract])) OR (consensus[Title/Abstract])) OR (best practice[Title/Abstract])
#11 #1 OR #2
#12 #3 OR #4
#13 #5 OR #6 OR #7
#14 #8 OR #9 OR #10
#15 #11 AND #12 AND #13 AND #14

For peer review only

Determination of eligibility

Two reviewers (LH and JX) will independently screen and select the literature. First, they will determine whether the documents meets the inclusion criteria based on the titles and abstracts. After that, the reviewers will obtains the full texts of the documents, then read and filter out the eligible guidelines. Any disagreement between the two reviewers will be resolved through consensus or by the third reviewer (ZZ).

DATA EXTRACTION

We will extract the following information of CPGs: (1) title; (2) year of publication; (3) authors; (4) country of development; (5) objective of CPG; (6) target users; (7) method used to collect the evidence; (8) grading methods for the evidence; (9) recommendation formulation method; (10) guideline validation methods; (11) intended users; (12) the specific content of the recommendations given, including the method of thromboprophylaxis, the timing of initiation, dose and duration, and target patients. In addition, we will record each score of items in AGREE-II and AGREE-REX with their reasons.

QUALITY ASSESSMENT OF CPGS

We will assess the quality of CPGs by AGREE-II [21]. The tool was released in 2009 and includes 23 items in six domains (*scope and purpose, stakeholder involvement, rigour of development, clarity of presentation, applicability, and editorial independence*), with two additional assessments for the guideline's overall quality and whether it is recommended to use.

The quality of recommendations is evaluated using AGREE-REX [22]. The tool was released in 2019 and included nine items in three domains (*clinical applicability, values and preferences, and implementability*). Each item has two assessment questions, one for quality evaluation (required) and another for suitability for use (optional). Overall judgment for all recommendations in CPG has two additional assessments: (1) *I would recommend these recommendations for use in the appropriate context* (required); (2) *I would recommend these recommendations for use in my context*

For peer review only

(optional).

Both AGREE-II and AGREE-REX quality evaluation items adopt a Likert scale of one to seven scores. The standardized scores for each domain are calculated according to the user manual [23,24], ranging from 0% to 100%. The higher the score obtained, the better the quality of the guideline in the relevant field. Considering that the method for evaluating the overall quality of the guideline remains controversial [25], we will not use a cut-off value to rate the overall quality but directly present the reviewers' subjective judgment results.

Five reviewers (LH, JX, JL, HL and NN) conducted quality appraisals after systematically learning how to use AGREE-II and AGREE-REX. Reviewers use the online platform *MY AGREE PLUS* to use AGREE-II independently and blindly (https://www.agreetrust.org/my-agree/), while the AGREE-REX assessment is performed offline.

DESCRIPTION AND COMPARISON OF THE RECOMMENDATIONS

We plan to compare the recommendations included in the CPG, which will be extracted independently by two reviewers (DL and GL). The recommendations will be expectantly classified according to risk assessment, screening strategies, mechanical prophylaxis, chemical prophylaxis, and other interventions to evaluate their scopes and differences in different CPGs. Discrepancies will be highlighted when there are conflicts between some recommendations.

DATA ANALYSIS AND PRESENTATION

All the data will be saved with Excel (Microsoft 365, Microsoft Corporation, Redmond, WA, USA) and analyzed with SPSS statistical software (Version 26, IBM Corporation, Armonk, NY, USA). For each domain of AGREE II and AGREE-REX, We will calculate the score by the following formula:

$$\textit{The scaled domain score} = \frac{\text{Obtained score} - \text{Minimum possible score}}{\text{Maximun possible score} - \text{Minimum possible score}}$$

Scores for guidelines and recommendations will be described using the mean (SD)

For peer review only

and median (IQR) (if needed). The intraclass correlation coefficient (ICC) will be used to evaluate the degree of agreement between reviewers. Based on Landis and Koch [26], an ICC value of 0.00 ~ 0.20 indicates slight agreement, 0.21 ~ 0.40 indicates fair agreement, 0.41 ~ 0.61 indicates moderate agreement, 0.61 ~ 0.80 indicates substantial agreement, 0.81 ~ 1.00 indicates almost perfect agreement. Differences in domain scores between CPGs will be compared using the Kruskal-Wallis H test. Significance level $\alpha = 0.05$ (two-tailed).

Our study is expected to begin in Nov 2022 and complete in April 2023. We will explain any methodological changes in the study. The results are intended to be published in a peer-reviewed open journal.

DISCUSSION

Thromboprophylaxis in orthopedic trauma patients is complicated due diverse risk factors. It is challenging to choose appropriate prevention methods for patients with coagulation abnormalities, lower limb injuries, etc. The use of anticoagulants may increase the risk of bleeding and wound complications in trauma patients [27–29], especially in the early postoperative period. Research in this field is scattered and controversial, and more work is needed to draw conclusions [30]. There is still no consensus on evaluating evidence and recommendations between different guidelines, which increases the uncertainty of clinical decision-making [31]. Even in the relatively large number of hip fracture patients studied, physicians still believe that there is a lack of adequate clinical guidelines [32]. We hope to evaluate the relevant guidelines for VTE prevention in orthopedic trauma and critically appraise the quality of the guidelines and recommendations. To the best of our knowledge, there is currently no systematic review on the quality of English and Chinese orthopedic trauma thromboprophylaxis guidelines.

It is a long process that costs plenty of workforce and resources for planning, formulating, and disseminating a CPG, especially for high-quality CPGs. Rigorous methodology and regular updates play essential roles in maintaining the quality of the guidelines and practical application [33]. Identifying high-quality recommendations

For peer review only

will help us to prescribe appropriate thromboprophylaxis in the trauma population. It will also help discover knowledge gaps and new points for further research [34]. We plan to summarize the divergent recommendations in such guidelines, which will be the main focus of future thromboprophylaxis research.

Our research may still have shortcomings. First, although we will adopted a comprehensive search strategy, only openly published documents will be obtained, and some gray literature may miss during the process. Second, we will only include CPGs published in Chinese and English, which may lead to an underestimation of the overall number of guidelines. Third, to obtain as comprehensive information as possible, we will not limit the start time for the publication of the guidelines. Those recommendations from earlier CPGs may not be consistent with current evidence.

CONTRIBUTORS

Study design: LH, JX, NN and ZZ. Protocol drafting: LX, JX, JL, HL and DL. Manuscript reviewing and editing: LX, GW, NN and ZZ. All the authors have read, discussed, and approved the final version of the manuscript. NN and ZZ contributed equally to the manuscript.

ACKNOWLEDGEMENTS

The authors gratefully acknowledge the suggestions given by Mr. Lei Luo on the manuscript language editing.

FUNDING

This research is supported by West China Nursing Discipline Development Special Fund Project, Sichuan University (No. HXHL20003).

COMPETING INTERESTS

None declared.

ETHICS AND DISSEMINATION

This study will follow the Declaration of Helsinki and has received approval from the Ethics Committee on Biomedical Research, West China Hospital, Sichuan University (Ethics Approval No. 2021-989). The results will be summarized as a paper, disseminated through peer-reviewed journals, and will help guide further research in

For peer review only

the future.

PATIENT CONSENT FOR PUBLICATION

Not required.

REFERENCES

1 Raskob GE, Angchaisuksiri P, Blanco AN, *et al.* Thrombosis: a major contributor to global disease burden. *Arterioscler Thromb Vasc Biol* 2014;**34**:2363–71. doi:10.1161/ATVBAHA.114.304488

2 Grosse SD, Nelson RE, Nyarko KA, *et al.* The economic burden of incident venous thromboembolism in the United States: A review of estimated attributable healthcare costs. *Thromb Res* 2016;**137**:3–10. doi:10.1016/j.thromres.2015.11.033

3 Zhan C. Excess Length of Stay, Charges, and Mortality Attributable to Medical Injuries During Hospitalization. *JAMA* 2003;**290**:1868. doi:10.1001/jama.290.14.1868

4 White RH. The epidemiology of venous thromboembolism. *Circulation* 2003;**107**:I4-8. doi:10.1161/01.CIR.0000078468.11849.66

5 Wolberg AS, Rosendaal FR, Weitz JI, *et al.* Venous thrombosis. *Nat Rev Dis Primers* 2015;**1**:15006. doi:10.1038/nrdp.2015.6

6 Wu L, Cheng B. Analysis of perioperative risk factors for deep vein thrombosis in patients with femoral and pelvic fractures. *J Orthop Surg Res* 2020;**15**:597. doi:10.1186/s13018-020-02131-5

7 Fan J, Zhou F, Xu X, *et al.* Clinical predictors for deep vein thrombosis on admission in patients with intertrochanteric fractures: a retrospective study. *BMC Musculoskelet Disord* 2021;**22**:328. doi:10/gkrb59

8 Kanchanabat B, Stapanavatr W, Meknavin S, *et al.* Systematic review and meta-analysis on the rate of postoperative venous thromboembolism in orthopaedic surgery in Asian patients without thromboprophylaxis. *Br J Surg* 2011;**98**:1356–64. doi:10.1002/bjs.7589

9 Yumoto T, Naito H, Yamakawa Y, *et al.* Venous thromboembolism in major trauma patients: a single-center retrospective cohort study of the epidemiology and utility of D-dimer for screening. *Acute Med Surg* 2017;**4**:394–400. doi:10.1002/ams2.290

10 Barrera LM, Perel P, Ker K, *et al.* Thromboprophylaxis for trauma patients.

For peer review only

Cochrane Database Syst Rev 2013;:CD008303. doi:10.1002/14651858.CD008303.pub2

11 Falck-Ytter Y, Francis CW, Johanson NA, *et al.* Prevention of VTE in Orthopedic Surgery Patients. Antithrombotic Therapy and Prevention of Thrombosis, 9th ed: American College of Chest Physicians Evidence-Based Clinical Practice Guidelines. *Chest* 2012;**141**:e278S-e325S. doi:10.1378/chest.11-2404

12 National Institute for Health and Care Excellence. Venous thromboembolism in over 16s: reducing the risk of hospital-acquired deep vein thrombosis or pulmonary embolism. National Institute for Health and Care Excellence. 2019.https://www.nice.org.uk/guidance/ng89 (accessed 27 Jun 2021).

13 Anderson DR, Morgano GP, Bennett C, *et al.* American Society of Hematology 2019 guidelines for management of venous thromboembolism: prevention of venous thromboembolism in surgical hospitalized patients. *Blood Adv* 2019;**3**:3898–944. doi:10.1182/bloodadvances.2019000975

14 Ngarmukos S, Kim K-I, Wongsak S, *et al.* Asia-Pacific venous thromboembolism consensus in knee and hip arthroplasty and hip fracture surgery: Part 1. Diagnosis and risk factors. *Knee Surg & Relat Res* 2021;**33**:18. doi:10.1186/s43019-021-00099-y

15 Amarase C, Tanavalee A, Larbpaiboonpong V, *et al.* Asia-Pacific venous thromboembolism consensus in knee and hip arthroplasty and hip fracture surgery: Part 2. Mechanical venous thromboembolism prophylaxis. *Knee Surg Relat Res* 2021;**33**:20. doi:10.1186/s43019-021-00101-7

16 Afshari A, Ageno W, Ahmed A, *et al.* European Guidelines on perioperative venous thromboembolism prophylaxis: Executive summary. *Eur J Anaesthesiol* 2018;**35**:77–83. doi:10.1097/EJA.0000000000000729

17 Aggarwal S, Patel S, Vashisht S, *et al.* Guidelines for the prevention of venous thromboembolism in hospitalized patients with pelvi-acetabular trauma. *Journal of Clinical Orthopaedics and Trauma* 2020;**11**:1002–8. doi:10.1016/j.jcot.2020.09.011

18 Lin Q, Yang M, Hou Z. Guidelines for prevention of perioperative venous thromboembolism in Chinese orthopedic trauma patients(2021). Chin J Orthop Trauma 2021;23:185–92. doi:10.3760/cma.j.cn115530-20201228-00795.

19 Ley EJ, Brown CVR, Moore EE, *et al.* Updated guidelines to reduce venous thromboembolism in trauma patients: A Western Trauma Association critical decisions algorithm. *J Trauma Acute Care Surg* 2020;**89**:971–81. doi:10.1097/TA.0000000000002830

20 Sagi HC, Ahn J, Ciesla D, *et al.* Venous Thromboembolism Prophylaxis in

For peer review only

Orthopaedic Trauma Patients: A Survey of OTA Member Practice Patterns and OTA Expert Panel Recommendations. *J Orthop Trauma* 2015;**29**:8. doi:10.1097/BOT.0000000000000387

21 Brouwers MC, Kho ME, Browman GP, *et al.* AGREE II: advancing guideline development, reporting and evaluation in health care. *CMAJ* 2010;**182**:E839-842. doi:10.1503/cmaj.090449

22 Brouwers MC, Spithoff K, Kerkvliet K, *et al.* Development and Validation of a Tool to Assess the Quality of Clinical Practice Guideline Recommendations. *JAMA Network Open* 2020;**3**:e205535–e205535. doi:10.1001/jamanetworkopen.2020.5535

23 AGREE Next Steps Consortium. The AGREE II Instrument. 2017.https://www.agreetrust.org/resource-centre/agree-ii/ (accessed 19 Oct 2021).

24 AGREE–REX Research Team. The Appraisal of Guidelines Research & Evaluation—Recommendation EXcellence (AGREE-REX). 2019.https://www.agreetrust.org/resource-centre/agree-rex-recommendation-excellence/ (accessed 19 Oct 2021).

25 Bargeri S, Iannicelli V, Castellini G, *et al.* AGREE II appraisals of clinical practice guidelines in rehabilitation showed poor reporting and moderate variability in quality ratings when users apply different cuff-offs: a methodological study. *Journal of Clinical Epidemiology* 2021;**139**:222–31. doi:10.1016/j.jclinepi.2021.08.021

26 Landis JR, Koch GG. The measurement of observer agreement for categorical data. *Biometrics* 1977;**33**:159–74.

27 Villarreal JV, Shibuya N, Jupiter DC. Thromboprophylaxis and Bleeding Complications in Orthopedic and Trauma Patients: A Systematic Review. *The Journal of Foot and Ankle Surgery* 2021;**60**:1014–22. doi:10.1053/j.jfas.2021.03.010

28 Heijboer RRO, Lubberts B, Guss D, *et al.* Venous Thromboembolism and Bleeding Adverse Events in Lower Leg, Ankle, and Foot Orthopaedic Surgery with and without Anticoagulants. *J Bone Joint Surg Am* 2019;**101**:539–46. doi:10.2106/JBJS.18.00346

29 Jones CW, Spasojevic S, Goh G, *et al.* Wound Discharge After Pharmacological Thromboprophylaxis in Lower Limb Arthroplasty. *J Arthroplasty* 2018;**33**:224–9. doi:10.1016/j.arth.2017.07.046

30 Flevas DA, Megaloikonomos PD, Dimopoulos L, *et al.* Thromboembolism prophylaxis in orthopaedics: an update. *EFORT Open Rev* 2018;**3**:136–48. doi:10.1302/2058-5241.3.170018

For peer review only

31 Gesu E, Bellone P, Bonzi M, *et al.* Management of patients with cervical spine trauma in the emergency department: a systematic critical appraisal of guidelines with a view to developing standardized strategies for clinical practice. *Intern Emerg Med* Published Online First: 5 October 2021. doi:10.1007/s11739-021-02838-1

32 White NJ, Reitzel SL, Doyle-Baker D, *et al.* Management of patients with hip fracture receiving anticoagulation: What are we doing in Canada? *Can J Surg* 2021;**64**:E510–5. doi:10.1503/cjs.018520

33 Cosgrove L, Shaughnessy AF, Shaneyfelt T. When is a guideline not a guideline? The devil is in the details. *BMJ Evidence-Based Medicine* 2018;**23**:33–6. doi:10.1136/ebmed-2017-110845

34 Murad MH. Clinical Practice Guidelines: A Primer on Development and Dissemination. *Mayo Clinic Proceedings* 2017;**92**:423–33. doi:10.1016/j.mayocp.2017.01.001

For peer review only

PRISMA-P 2015 Checklist

This checklist has been adapted for use with protocol submissions to *Systematic Reviews* from Table 3 in Moher D et al: Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Systematic Reviews* 2015 **4**:1

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
ADMINISTRATIVE INFORMATION					
Title					
Identification	1a	Identify the report as a protocol of a systematic review	☒	☐	1-3
Update	1b	If the protocol is for an update of a previous systematic review, identify as such	☐	☒	n/a
Registration	2	If registered, provide the name of the registry (e.g., PROSPERO) and registration number in the Abstract	☒	☐	41
Authors					
Contact	3a	Provide name, institutional affiliation, and e-mail address of all protocol authors; provide physical mailing address of corresponding author	☒	☐	online
Contributions	3b	Describe contributions of protocol authors and identify the guarantor of the review	☒	☐	255-258
Amendments	4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	☐	☒	n/a
Support					
Sources	5a	Indicate sources of financial or other support for the review	☒	☐	262-263
Sponsor	5b	Provide name for the review funder and/or sponsor	☒	☐	262-263
Role of sponsor/funder	5c	Describe roles of funder(s), sponsor(s), and/or institution(s), if any, in developing the protocol	☒	☐	262-263
INTRODUCTION					
Rationale	6	Describe the rationale for the review in the context of what is already known	☒	☐	60-98
Objectives	7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	☒	☐	101-110

For peer review only

BioMed Central The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
METHODS					
Eligibility criteria	8	Specify the study characteristics (e.g., PICO, study design, setting, time frame) and report characteristics (e.g., years considered, language, publication status) to be used as criteria for eligibility for the review	☒	☐	121-124
Information sources	9	Describe all intended information sources (e.g., electronic databases, contact with study authors, trial registers, or other grey literature sources) with planned dates of coverage	☒	☐	127-141
Search strategy	10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	☒	☐	143-147
STUDY RECORDS					
Data management	11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	☒	☐	203-219
Selection process	11b	State the process that will be used for selecting studies (e.g., two independent reviewers) through each phase of the review (i.e., screening, eligibility, and inclusion in meta-analysis)	☒	☐	149-154
Data collection process	11c	Describe planned method of extracting data from reports (e.g., piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	☒	☐	156-164
Data items	12	List and define all variables for which data will be sought (e.g., PICO items, funding sources), any pre-planned data assumptions and simplifications	☒	☐	156-164
Outcomes and prioritization	13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	☒	☐	166-171
Risk of bias in individual studies	14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	☒	☐	172–186
DATA					
Synthesis	15a	Describe criteria under which study data will be quantitatively synthesized	☐	☒	n/a
	15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data, and methods of combining data from studies, including any planned exploration of consistency (e.g., I^2, Kendall's tau)	☐	☒	n/a
	15c	Describe any proposed additional analyses (e.g., sensitivity or subgroup analyses, meta-regression)	☐	☒	n/a
	15d	If quantitative synthesis is not appropriate, describe the type of summary planned	☒	☐	193-201
Meta-bias(es)	16	Specify any planned assessment of meta-bias(es) (e.g., publication bias across studies, selective	☐	☒	n/a

For peer review only

BioMed Central
The Open Access Publisher

Section/topic	#	Checklist item	Information reported		Line number(s)
			Yes	No	
		reporting within studies)			
Confidence in cumulative evidence	17	Describe how the strength of the body of evidence will be assessed (e.g., GRADE)	☐	☒	n/a

For peer review only

BioMed Central The Open Access Publisher

Table S1. Search strategy for databases, registries and websites

Database	Search terms
PubMed	#1 ((orthopedics[MeSH Terms]) OR (traumatology[MeSH Terms])) OR (wounds and injuries[MeSH Terms]) #2 (((((((fracture[Title/Abstract]) OR (limb trauma[Title/Abstract])) OR (extremity trauma[Title/Abstract])) OR (musculoskeletal injur*[Title/Abstract])) OR (musculoskeletal trauma[Title/Abstract])) OR (limb immobili*[Title/Abstract])) OR (skeletal fixation[Title/Abstract])) OR ("orthop*"[Title/Abstract] AND "trauma"[Title/Abstract]) #3 ((venous thromboembolism[MeSH Terms]) OR (venous thrombosis[MeSH Terms])) OR (pulmonary embolism[MeSH Terms]) #4 ((((((((((venous thromboembolism[Title/Abstract]) OR (venothromboembolism[Title/Abstract])) OR (venous thrombosis[Title/Abstract])) OR (vein thrombosis[Title/Abstract])) OR (DVT[Title/Abstract])) OR (VTE[Title/Abstract])) OR (pulmonary embolism[Title/Abstract])) OR (lung embolism[Title/Abstract])) OR (PE[Title/Abstract])) OR (thromboembolism[Title/Abstract])) OR (embolism[Title/Abstract])) OR (blood clot[Title/Abstract]) #5 ((thromboprophylaxis[Title/Abstract]) OR (prophyla*[Title/Abstract])) OR (prevent*[Title/Abstract]) #6 (((((((heparin[MeSH Terms]) OR (heparin, low-molecular-weight[MeSH Terms])) OR (anticoagulants[MeSH Terms])) OR (warfarin[MeSH Terms])) OR (aspirin[MeSH Terms])) OR (vena cava filters[MeSH Terms])) OR (stockings, compression[MeSH Terms])) OR (intermittent pneumatic compression devices[MeSH Terms]) #7 ((pentasaccharide*[Title/Abstract]) OR (fondaparinux[Title/Abstract])) OR (LMWH[Title/Abstract]) #8 (guideline[Publication Type]) OR (consensus development conference[Publication Type]) #9 ((guidelines as topic[MeSH Terms]) OR (consensus[MeSH Terms])) OR (consensus development conferences as topic[MeSH Terms]) #10 ((((guideline[Title/Abstract]) OR (guidance[Title/Abstract])) OR (recommendation[Title/Abstract])) OR (consensus[Title/Abstract])) OR (best practice[Title/Abstract]) #11 #1 OR #2 #12 #3 OR #4 #13 #5 OR #6 OR #7 #14 #8 OR #9 OR #10 #15 #11 AND #12 AND #13 AND #14
	1 exp orthopedics/ 2 exp traumatology/ 3 exp musculoskeletal injury/ 4 fracture.ab,ti. 5 limb trauma.ab,ti. 6 extremity trauma.ab,ti. 7 musculoskeletal injur$.ab,ti. 8 musculoskeletal trauma.ab,ti. 9 limb immobili$.ab,ti. 10 skeletal fixation.ab,ti. 11 (orthop$ adj3 trauma).ab,ti. 12 exp venous thromboembolism/ 13 exp lung embolism/ 14 venous thromboembolism.ab,ti. 15 venothromboembolism.ab,ti.

For peer review only

	16 venous thrombosis.ab,ti. 17 vein thrombosis.ab,ti. 18 DVT.ab,ti. 19 VTE.ab,ti. 20 pulmonary embolism.ab,ti. 21 lung embolism.ab,ti. 22 PE.ab,ti. 23 thromboembolism.ab,ti. 24 embolism.ab,ti. 25 blood clot.ab,ti. 26 exp heparin/ or exp low molecular weight heparin/ 27 exp anticoagulant agent/ 28 exp warfarin/ 29 exp acetylsalicylic acid/ 30 exp vena cava filter/ 31 exp compression stocking/ 32 exp intermittent pneumatic compression device/ 33 thromboprophylaxis.ab,ti. 34 prophyla$.ab,ti. 35 prevent$.ab,ti. 36 pentasaccharide$.ab,ti. 37 fondaparinux.ab,ti. 38 LMWH.ab,ti. 39 exp practice guideline/ 40 exp consensus/ or exp consensus development/ 41 guideline.ab,ti. 42 guidance.ab,ti. 43 recommendation.ab,ti. 44 consensus.ab,ti. 45 best practice.ab,ti. 46 1 or 2 or 3 or 4 or 5 or 7 or 8 or 9 or 10 or 11 47 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 48 26 or 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 49 39 or 40 or 41 or 42 or 43 or 44 or 45 50 46 and 47 and 48 and 49
EBSCO-CINAHL	S1 MH Orthopedic Surgery OR MH Trauma OR MH (Wounds and Injuries) S2 AB fracture OR AB limb trauma OR AB extremity trauma OR AB musculoskeletal injur* OR AB musculoskeletal trauma OR AB limb immobili* OR AB skeletal fixation OR AB orthop#dic trauma S3 MH Venous Thromboembolism OR MH Pulmonary Embolism S4 AB venous thromboembolism OR AB venothromboembolism OR AB venous thrombosis OR AB vein thrombosis OR AB (DVT OR VTE OR PE) OR AB pulmonary embolism OR AB lung embolism OR AB thromboembolism OR AB embolism OR AB blood clot S5 MH Heparin, Low-Molecular-Weight OR MH Heparin OR MH Anticoagulants OR MH Warfarin OR MH Aspirin OR MH Vena Cava Filters OR MH Compression Garments S6 AB thromboprophylaxis OR AB prophyla* OR AB prevent* OR AB pentassacharide* OR AB fondaparinux OR AB compression stocking* OR AB intermittent pneumatic compression S7 MH Practice Guidelines OR MH Consensus S8 AB guideline OR AB guidance OR AB consensus OR AB recommendation OR AB best practice

For peer review only

	S9 S1 OR S2 S10 S3 OR S4 S11 S5 OR S6 S12 S7 OR S8 S13 S9 AND S10 AND S11 AND S12
Web of Science	1 TS=(fracture OR "limb trauma" OR "extremity trauma" OR "musculoskeletal injur*" OR "musculoskeletal trauma" OR "limb immobili*" OR "skeletal fixation" OR "orthop* NEAR/3 trauma") 2 TS=("venous thromboembolism" OR venothromboembolism OR "venous thrombosis" OR "vein thrombosis" OR DVT OR VTE OR "pulmonary embolism" OR "lung embolism" OR PE OR thromboembolism OR embolism OR "blood clot") 3 TS=(thromboprophylaxis OR prophyla* OR prevent* OR heparin OR LMWH OR anticoagulants OR warfarin OR aspirin OR "vena cava filters" OR "compression stocking" OR "intermittent pneumatic compression devices" OR IPC OR pentassacharide* OR fondaparinux) 4 TS=(guideline OR guidance OR consensus OR recommendation OR "best practice") 5 #4 AND #3 AND #2 AND #1
Cochrane Library(via Ovid)	1 exp orthopedics/ 2 exp traumatology/ 3 exp "wounds and injuries"/ 4 fracture.ab,ti. 5 limb trauma.ab,ti. 6 extremity trauma.ab,ti. 7 musculoskeletal injur$.ab,ti. 8 musculoskeletal trauma.ab,ti. 9 limb immobili$.ab,ti. 10 skeletal fixation.ab,ti. 11 (orthop$ adj3 trauma).ab,ti. 12 exp venous thromboembolism/ 13 exp venous thrombosis/ 14 exp pulmonary embolism/ 15 venous thromboembolism.ab,ti. 16 venothromboembolism.ab,ti. 17 venous thrombosis.ab,ti. 18 vein thrombosis.ab,ti. 19 DVT.ab,ti. 20 VTE.ab,ti. 21 pulmonary embolism.ab,ti. 22 lung embolism.ab,ti. 23 PE.ab,ti. 24 thromboembolism.ab,ti. 25 embolism.ab,ti. 26 blood clot.ab,ti. 27 exp heparin/ 28 exp heparin, low-molecular-weight/ 29 exp anticoagulants/ 30 exp warfarin/ 31 exp aspirin/ 32 exp vena cava filters/ 33 exp stockings, compression/ 34 exp intermittent pneumatic compression devices/ 35 thromboprophylaxis.ab,ti. 36 prophyla$.ab,ti.

For peer review only

	37 prevent$.ab,ti. 38 pentassacharide$.ab,ti. 39 fondaparinux.ab,ti. 40 LMWH.ab,ti. 41 exp guidelines as topic/ 42 exp consensus development conferences as topic/ 43 exp consensus/ 44 guideline.ab,ti. 45 guidance.ab,ti. 46 recommendation.ab,ti. 47 consensus.ab,ti. 48 best practice.ab,ti. 49 1 or 2 or 3 or 4 or 5 or 6 or 7 or 8 or 9 or 10 or 11 50 12 or 13 or 14 or 15 or 16 or 17 or 18 or 19 or 20 or 21 or 22 or 23 or 24 or 25 or 26 51 27 or 28 or 29 or 30 or 31 or 32 or 33 or 34 or 35 or 36 or 37 or 38 or 39 or 40 52 41 or 42 or 43 or 44 or 45 or 46 or 47 or 48 53 49 and 50 and 51 and 52
SinoMed	1 (("矫形外科学"[加权:扩展] OR "创伤和损伤"[加权:扩展] OR "创伤学"[加权:扩展]) OR("骨折"[常用字段] OR "创伤"[常用字段] OR "肌肉骨骼损伤"[常用字段] OR "肢体制动"[常用字段] OR "内固定"[常用字段] OR "外固定"[常用字段])) AND (人类[特征词]) 2 (("静脉血栓形成"[加权:扩展] OR "静脉血栓栓塞"[加权:扩展] OR "肺栓塞"[加权:扩展]) OR("静脉血栓"[常用字段] OR "肺栓塞"[常用字段] OR "肺梗死"[常用字段] OR "DVT"[常用字段] OR "VTE"[常用字段] OR "PE"[常用字段])) AND (人类[特征词]) 3 (("肝素,低分子量"[加权:扩展] OR "阿司匹林"[加权:扩展] OR "华法林"[加权:扩展] OR "抗凝药"[加权:扩展] OR "化学预防"[加权:扩展] OR "腔静脉滤器"[加权:扩展] OR "间歇性气体压缩装置"[加权:扩展] OR "长袜,压力"[加权:扩展]) OR("化学预防"[常用字段] OR "血栓预防"[常用字段] OR "机械预防"[常用字段] OR "弹力袜"[常用字段] OR "间歇加压装置"[常用字段] OR "LMWH"[常用字段] OR "戊糖"[常用字段] OR "磺达肝癸"[常用字段])) AND (人类[特征词]) 4 (("准则"[加权:扩展] OR "多数赞同"[加权:扩展] OR "总结性报告"[加权:扩展]) OR("指南"[常用字段] OR "共识"[常用字段] OR "规范"[常用字段] OR "指引"[常用字段] OR "专家意见"[常用字段])) AND (人类[特征词]) 5 (#1) AND (#2) AND (#3) AND (#4)
CNKI	SU=‘骨折 + 骨科 + 创伤 + 肌肉骨骼损伤 + 内固定 + 外固定’ AND SU=‘静脉血栓 + 肺栓塞 + 肺梗死 + DVT + VTE + PE’ AND SU=‘肝素 + 低分子肝素 + 阿司匹林 + 华法林 + 抗凝药 + 血栓预防 + 化学预防 + 机械预防 + 腔静脉滤器 + 间歇充气加压 + 弹力袜 + 间歇加压装置 + LMWH + 戊糖 + 磺达肝癸’ AND SU=‘指南 + 共识 + 指引 + 专家意见 + 规范’
WanFang	主题:(骨折 OR 骨科 OR 创伤 OR 肌肉骨骼损伤 OR 内固定 OR 外固定) and 主题:(静脉血栓 OR 肺栓塞 OR 肺梗死 OR DVT OR VTE OR PE) and 主题:(肝素 OR 低分子肝素 OR 阿司匹林 OR 华法林 OR 抗凝药 OR 血栓预防 OR 化学预防 OR 机械预防 OR 腔静脉滤器 OR 间歇充气加压 OR 弹力袜 OR LMWH OR 戊糖 OR 磺达肝癸) and 主题:(指南 OR 指引 OR 共识 OR 专家意见 OR 规范)
Academic Websites	thromboembolism OR deep vein thrombosis OR pulmonary embolism OR VTE OR 静脉血栓 OR 肺栓塞 OR 肺梗死

For peer review only

Guideline repositories	thromboembolism OR deep vein thrombosis OR pulmonary embolism OR VTE OR 静脉血栓 OR 肺栓塞 OR 肺梗死

For peer review only